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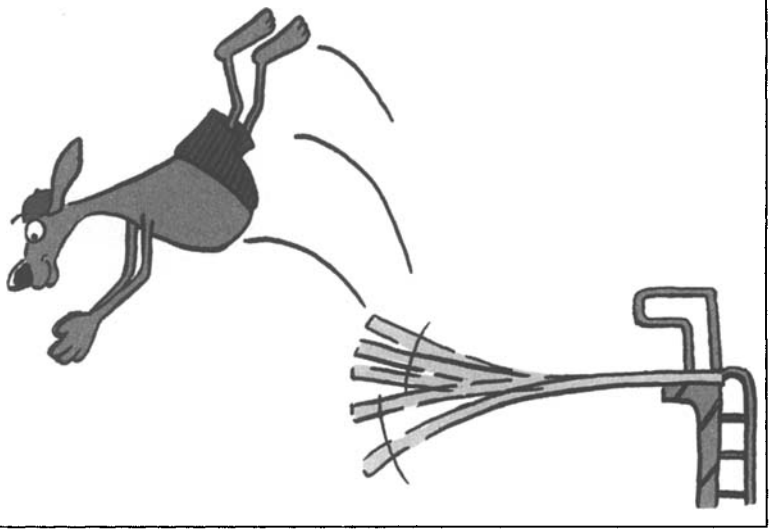
5

Distributed Systems



5.1 INTRODUCTION

Up until now the systems we've been examining have been composed of easily identified mass and spring elements. The masses and springs always appeared discretely, never in a distributed or continuous manner. But in the real world, such idealized structures never exist. The fact of the matter is that real-world structures are *always* continuous. Nobody can construct a true point mass; all real bodies have finite dimensions and so are characterized by their density or mass per unit volume. Similarly, ideal springs exist only in our minds. In the simple experiments we conducted in Chapters 1, 2, and 4, we treated the physical spring as a pure spring. But it is clear that any physical spring always has mass, a mass that we neglected in our analysis.

So, if we can never construct a system made up of just masses and springs, why did we study them? In the first place, although we can never have systems whose mass and spring elements occur in a precisely discrete manner, we can often approximate real systems as being discrete with only a negligible loss in accuracy. In addition, the mathematics associated with discrete systems is a bit easier to deal with. And finally, we'll be introducing methods that explicitly include continuous responses, yet put the overall system into a discrete framework.

The point is that we may find ourselves wanting to analyze realistic systems having continuous distributions of mass, elasticity, and damping. The problem is that except in some special cases, we cannot solve continuous problems exactly. Thus we'll find ourselves forced to consider approximate solutions. And we'll see that the approximate solutions fall into two distinct classes: those that discretize the original system into a number of discrete mass and stiffness elements (surprise—back to lumped elements!) and those that approximate the system's response by a finite number of mode shapes. We've already seen that MDOF systems possess a finite number of modes, i.e., free vibration configurations of the mass displacements that occur at the system's natural frequencies. What we'll see in this chapter is that continuous systems also have modes, but in this case they have an infinite number of them. It so happens that the modal approach (that of approximating the system's response through a finite number of mode shapes) is very widely used and forms the basis of modal analysis, the topic of Chapter 8.

Although there are several types of system that can be solved exactly, we'll look at only two of them in this chapter and leave some others to Appendix A (which interested readers can peruse at their leisure). The underlying mathematical analysis is exactly the same for all these problems, hence once you've solved one you can pretty much solve the rest as well. The two examples presented in this chapter are longitudinal vibrations in a bar and transverse vibrations of a beam. The bar example will also be used later (in Chapter 6) to illustrate how one can approximate a continuous system by a lumped one. What's interesting is that in their mathematical form, the bar equations are *exactly* the same as those that govern the vibrations of a tensioned string, which are identical to those that govern the torsional motions of a circular rod, which happen to be precisely the same as those governing the behavior of acoustic waves in a one-dimensional duct. Thus, once you've solved one of these problems, you've solved them all—the only things that change are the meaning of the dependent variable (transverse deflection of a string, rotation of the circular rod, etc.) and the meaning of the particular physical quantities (tension in the string, torsional stiffness of the rod, etc.).

We'll also be looking at transverse motions of a beam because this problem is very, very common in practical applications—almost all structures contain beams or beamlike elements. Since you'll be practicing engineering in the real world, it'll be helpful to have some knowledge of beam behavior. It'd be nice if the beam equations were also exactly the same as those of the string, rod, bar, etc., but unfortunately they're not. Luckily, the differences aren't too bad, as we'll soon see.

Before we get involved with some actual continuous systems, you should think about what the differences are between a vector and a function, since these are the relevant quantities for discrete and continuous problems, respectively. Let's consider position as the variable that we're measuring. In this case, a vector is simply a string of numbers. The position of the numbers within the vector tell us where on the structure we are, and the numbers themselves tell us how much motion is taking place at a particular position.

Take, for example, the deflection of a string. A simple way to represent the string's position would be to use three pieces of data: the end deflections and the deflection in the middle. A result for

a string that's tied down on both its ends would be

$$\begin{Bmatrix} 0 \\ 1 \\ 0 \end{Bmatrix} \tag{5.1.1}$$

This vector tells us that the leftmost end's deflection (the first entry) is zero, the middle deflection is 1 (the second entry) and the right end's deflection (the third entry) is zero. This is depicted in Figure 5.1, where the middle deflection is taken to be 1 cm and the string's length is taken to be 1 m.

We can now ask ourselves what the difference would be if, instead of having only 3 positions along the string, we had 101. If these were again uniformly distributed, then the second entry in (5.1.1) would become the 51st entry in our new position vector. It would still be the middle entry, but we'd now have many additional entries. A plot of the new vector is shown in Figure 5.2. You'll note that we've got a lot more data about what's going on at other points, but our knowledge of what's happening at $x = .5$ m is the same as it was. If we had an interest in the behavior at $x = .25$ m and were only using the 3-entry vector, we'd have had to estimate the answer by using the first and second entries and averaging them, giving us an estimate of .5 cm. With the greater precision that the 101-entry vector gives us, we can simply read the result off as the value associated with the 25th entry of the vector: .707 cm.

You'll notice that the plot of the vector shown in Figure 5.2 looks an awful lot like half of a sine wave. If we used a 1001-entry vector instead of an 101-entry one, the plot would look like that shown in Figure 5.3. This is now virtually indistinguishable from half a sine. What's happening here? The answer is that our vector is "trying" to become a function. As we add more and more entries, it "looks" closer and closer to a continuous function. That's all a function really is! It's a very long vector. In fact, it's a vector in which the number of entries has gone to infinity. Pretty neat, eh? This means that,

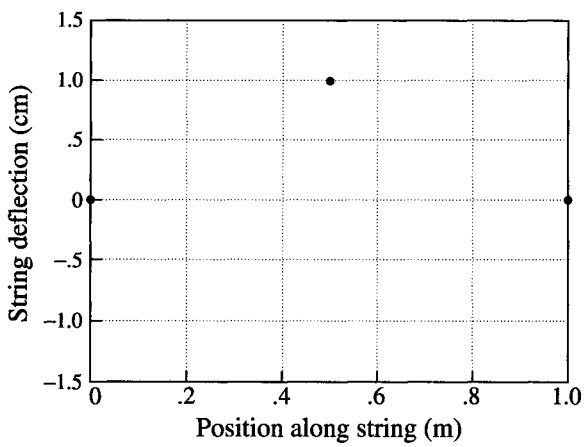


Figure 5.1 Pictorial representation of a 3-element vector

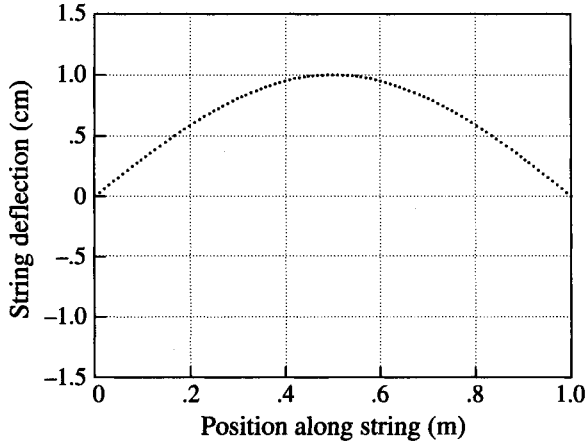


Figure 5.2 Pictorial representation of a 101-element vector

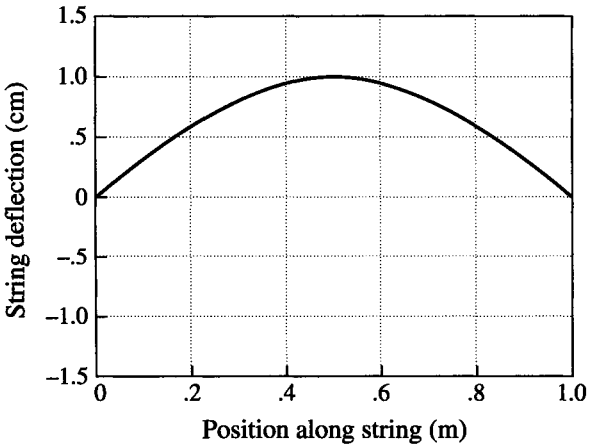


Figure 5.3 Pictorial representation of a 1001-element vector

as far as our computers are concerned, there is *no* difference between vectors and functions, since the computers can only deal with finite strings of information. Mathematically there's a difference (since mathematics has no problems with infinity), but practically we'll find that large vectors and functions are handled in very similar ways.

You should keep this little demonstration in mind as you encounter continuous systems in this and following chapters. All the continuous operations can really be viewed as discrete operations in which all the operations go to their limits. For example, summation of vector entries will mutate into integrations as we go from discrete to continuous descriptions. The distance between two entries of a position vector will go to a differential distance dx as the vector becomes bigger and bigger and finally becomes a function. If you have a problem following what's being done on the continuous problem, try thinking about it in a discrete form and see if that makes it clearer.

5.2 FREE VIBRATION OF A BAR (ROD, STRING, ETC.)

As the heading says, this section looks at bar vibrations, among other applications. Appendix A shows the equations of motion for several systems, all of which are identical in form although different in the specific parameters used. We'll perform a full derivation of the equations of motion only for a bar, realizing that the approach for all the other applications is essentially the same.

To begin studying bar vibrations, we must derive the equations of motion for our system. The bar is of length l and we identify locations along the bar with the coordinate x (Figure 5.4). Since the bar is deformable, individual elements will move from their initial positions if a force is applied. The amount of movement will be different at different points along the bar, and so we should use $\xi(x, t)$ to indicate these deformations. However, this would quickly become tiring because we'd have to write a lot of $\xi(x, t)$'s in the coming pages. Thus we'll simply write ξ. Just keep in mind that this refers to a function that is both position and time dependent. Also, be clear that the bar's motion is horizontal in this case; it's motion *along* the bar (stretching and compressing of the bar).

Please make sure the distinction between x and ξ is clear in your mind. x indicates a position *on the bar*. If you were asked to put a pencil mark on the bar 5 cm from the left, then the pencil mark is at $x = 5$ cm. $\xi(x, t)$ indicates *how much that pencil mark moves* if the bar is vibrating. The x and t dependences indicate that the deformation depends on the location along the bar as well as on time.

In your deformable bodies course you learned that the relation between applied an force, f, and the resultant deformation of the bar is given by

$$f = EA(x)\xi_x \tag{5.2.1}$$

where E is Young's modulus and $A(x)$ is the bar's cross-sectional area. This result means that the spatial rate of change of deformation is proportional to the applied force.

If we separate a small element from the bar, as shown in Figure 5.5, and assume that the motion occurs purely in the longitudinal direction, then we can find the equation of motion by balancing the inertial and applied forces:

$$m(x)\ddot{\xi}dx = [EA(x) + (EA(x))_x dx]\,[\xi_x + \xi_{xx}dx] - EA(x)\xi_x \tag{5.2.2}$$

Note that $m(x)$ denotes the mass per unit length. Multiplying this equation out and canceling the $EA(x)\xi_x$ terms yields

$$m(x)\ddot{\xi}dx = (EA(x)_x)\,\xi_x dx + EA(x)\xi_{xx}dx + (EA(x))_x\,\xi_{xx}(dx)^2 \tag{5.2.3}$$

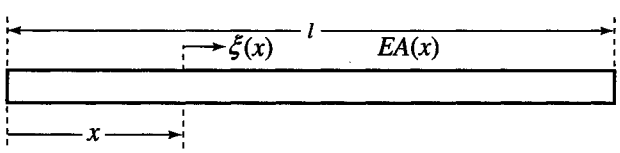


Figure 5.4 Schematic of a bar

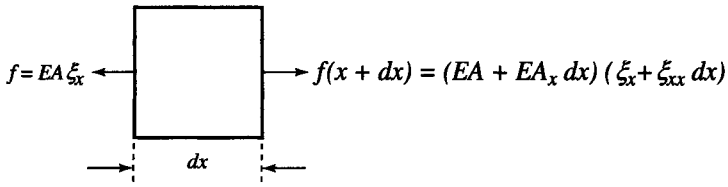


Figure 5.5 Differential element of a bar

We've got a common factor of dx, and canceling it out leaves us with

$$m(x)\ddot{\xi} = (EA(x)_x)\,\xi_x + EA(x)\xi_{xx} + (EA(x))_x\,\xi_{xx}dx \tag{5.2.4}$$

Keep in mind that the analysis we are doing is an infinitesimal one; i.e., the element we've taken out of the bar is of only infinitesimal length. If we take the limit of dx (the length of the element) going to zero, then the $(EA(x))_x\,\xi_{xx}dx$ term will go to zero. Thus we can eliminate this term, and we are left with

$$m(x)\ddot{\xi} = EA(x)_x\xi_x + EA(x)\xi_{xx} \tag{5.2.5}$$

which can be written more compactly as

$$m(x)\ddot{\xi} = \frac{\partial}{\partial x}\,(EA(x)\xi_x) \tag{5.2.6}$$

To keep the analysis simple, we will assume that $m(x)$ and $EA(x)$ are constant. Thus, even though in general they might vary along the beam, in our case they do not. The basic reason for using this simplifying assumption is that letting these terms vary along the beam moves this from the "simple" category and into the "just about impossible" category. Luckily, Chapter 6 shows how to deal with these more difficult cases. Note that the second term of (5.2.5) disappears under this assumption, since the spatial variation has been set to zero.

Dropping the x dependence from our notation and using the fact that EA and m are constant gives us

$$m\ddot{\xi} = EA\xi_{xx} \tag{5.2.7}$$

Dividing by m serves to group all the material parameters together and leaves us with

$$\ddot{\xi} = \frac{EA}{m}\xi_{xx} \tag{5.2.8}$$

At this point we need to invoke a new mathematical technique to get any further. The technique we'll use is called the *separation of variables* method. You may have seen this approach in a math class and it may have seemed a bit obscure. Hopefully, this example will clear up any confusion.

Simply put, the method requires us to assume that $\xi(x, t)$ can be written in the form

$$\xi(x, t) = \mathbf{x}(x)\mathbf{t}(t) \tag{5.2.9}$$

where $\mathbf{x}(x)$ and $\mathbf{t}(t)$ are as yet undetermined functions of position and time. Although this is often a very reasonable assumption, it should be stressed that it is *only an assumption* at this point. We'll

have to carry on and see if we eventually get a reasonable answer. If we do, then we'll know the assumption was justified. As a quick demonstration that this assumption doesn't always have to be satisfied, just try and put $xt + x^2$ into this form.

Substituting (5.2.9) into (5.2.8) yields

$$\frac{EA}{m}\mathbf{x}_{xx}\mathbf{t} = \mathbf{x}\ddot{\mathbf{t}} \tag{5.2.10}$$

If we divide by $\mathbf{xt}$, we'll obtain

$$\frac{EA}{m}\frac{\mathbf{x}_{xx}}{\mathbf{x}} = \frac{\ddot{\mathbf{t}}}{\mathbf{t}} \tag{5.2.11}$$

From this result we see that the term on the left (which only depends on position) has to be equal to the term on the right (which only depends on time). This is the point that is invariably the most confusing. Ask yourself how two quantities can be equal when each depends upon a different dependent variable. Clearly, this is a tough one. For instance, the term on the left might conceivably be equal to $\sin(2x)$ and the term on the right equal to t^3. But then they *can't* be equal to each other for arbitrarily varying values of x or t. In fact, functions that depend on completely different variables can equal each other only if they *don't vary at all*—if they're equal to a constant. Thus we need to require that

$$\frac{EA}{m}\frac{\mathbf{x}_{xx}}{\mathbf{x}} = c \tag{5.2.12}$$

and

$$\frac{\ddot{\mathbf{t}}}{\mathbf{t}} = c \tag{5.2.13}$$

as well.

Taking (5.2.13) first, we multiply by $\mathbf{t}$ and pull both terms to the same side of the equation

$$\ddot{\mathbf{t}} - c\mathbf{t} = 0 \tag{5.2.14}$$

At this point it makes sense to replace c with $-\omega^2$. Why? Well, think about it for a moment. The system we're looking at has spring and mass properties. We've already seen what springs and masses do when perturbed from equilibrium—they oscillate. There's no logical reason to expect the bar to behave any differently. In fact, one way to think of a bar is simply as a collection of many small masses connected to many small springs. If we use the particular form $-\omega^2$ for our constant, then (5.2.14) will be in the form of an oscillator equation, exactly analogous to the oscillators we first saw in Chapter 1.

Replacing c with $-\omega^2$ gives us

$$\ddot{\mathbf{t}} + \omega^2\mathbf{t} = 0 \tag{5.2.15}$$

which, as we know, has solutions

$$\mathbf{t}(t) = a_1\cos(\omega t) + a_2\sin(\omega t) \tag{5.2.16}$$

What this tells us is that the bar undergoes sinusoidal motion at a frequency ω. As far as we know right now, any ω is as good as any other; there's nothing to indicate that the value of ω isn't completely arbitrary. However, as we'll see next, the actual choices for ω are quite restricted and are closely tied to the physical configuration of the system.

Since there's nothing more to be learned from (5.2.13), we'll move on to (5.2.12). Once we've substituted $-\omega^2$ for c, we see that multiplying (5.2.12) by $\frac{m}{EA}\mathbf{x}$ and pulling all terms to the left-hand side leaves us with

$$\mathbf{x}_{xx} + \frac{m\omega^2}{EA}\mathbf{x} = 0 \tag{5.2.17}$$

Note that perhaps surprisingly, this is in the form of an oscillator equation. So, not only does the bar oscillate in time, it now appears that it oscillates in space as well. Very interesting. Just to keep the notation simple, let's define a new variable, β,

$$\beta \equiv \omega\sqrt{\frac{m}{EA}} \tag{5.2.18}$$

so that (5.2.17) becomes

$$\ddot{\mathbf{x}}_{xx} + \beta^2\mathbf{x} = 0 \tag{5.2.19}$$

with solutions

$$\mathbf{x}(x) = b_1\cos(\beta x) + b_2\sin(\beta x) \tag{5.2.20}$$

At this point we seem to be stuck. We have to determine what β is equal to (which will tell us ω from (5.2.18)), but we don't seem to have enough information yet to do this. This is because we can't find the precise form of the solution until we've fully pinned down the physics of the problem. And we haven't done that yet. Perhaps you've already noticed that no mention has been made of the boundary conditions at the bar ends. Is the rod fixed to a wall? Is it free to move around? Until we decide on what's happening at the bar ends, we'll be unable to progress further. And this makes sense, since it seems obvious that the bar's response should be different for different boundary conditions. So let's pick fixed-fixed conditions, the situation we'd encounter if the rod was built into rigid walls on both ends (Figure 5.6). In this case the ends can't move longitudinally and so

$$\xi(0, t) = 0 \tag{5.2.21}$$

and

$$\xi(l, t) = 0 \tag{5.2.22}$$

Since $\xi(x, t) = \mathbf{x}(x)\mathbf{t}(t)$, we see that these boundary conditions imply that

$$\mathbf{x}(0) = \mathbf{x}(l) = 0 \tag{5.2.23}$$

Evaluating $\mathbf{x}(x)$ at $x = 0$ leaves us with $\mathbf{x}(0) = b_1$, since the sine term goes to zero and the cosine to 1. Since our boundary condition states that $\mathbf{x}(0) = 0$, we see that b_1 must be zero. Therefore

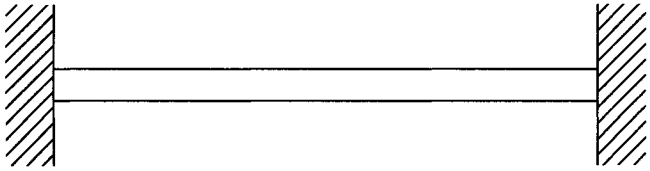


Figure 5.6 Bar with restrained ends

we're left with

$$\mathbf{x}(x) = b_2 \sin(\beta x) \tag{5.2.24}$$

Evaluating this expression at $x = l$ and using the remaining boundary condition yields

$$\mathbf{x}(l) = b_2 \sin(\beta l) = 0 \tag{5.2.25}$$

There are two ways (5.2.25) can be satisfied. The first is to set b_2 to zero. The only problem with this is that this would make $\mathbf{x}$ equal to zero, and so $\xi(x, t)$ would equal zero. This is clearly the trivial solution—no motion at all. Not very exciting. The other way in which (5.2.25) can be satisfied is for $\sin(\beta l)$ to equal zero. If we look at a plot of $\sin(\beta l)$ versus βl (Figure 5.7), we see that this condition occurs for an infinite number of nonzero β values. If we identify these values as $\beta_1, \beta_2, \ldots,$ we have

$$\beta_n = \frac{n\pi}{l}, \quad n = 1, 2, \ldots, \infty \tag{5.2.26}$$

Thus our spatial solution is

$$\mathbf{x}(x) = b_2 \sin(\beta_n x) \tag{5.2.27}$$

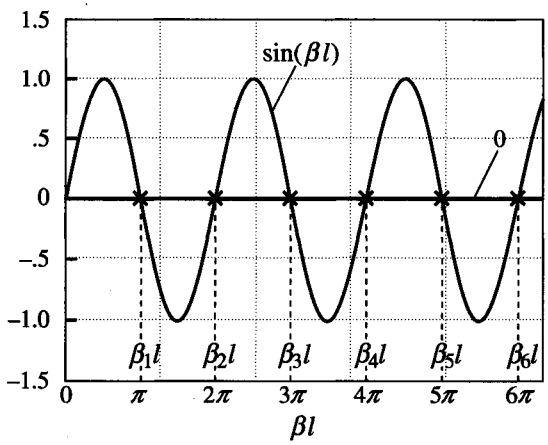


Figure 5.7 $\beta_n l$ values for a fixed-fixed bar

If we recall that $\beta = \omega\sqrt{\frac{m}{EA}}$, we see that we can now find the possible frequencies of oscillation for our system, i.e.,

$$\omega_n = \beta_n\sqrt{\frac{EA}{m}} \tag{5.2.28}$$

We have an infinite set of β_n values and an associated infinite set of ω_n's. We'll call $\sin(\beta_n x)$ the modes of our system and n the mode number.

Let's examine some of these solutions more closely. The first value of β (represented as β_1) is $\frac{\pi}{l}$. The mode shape associated with this β value is

$$\gamma(x) = \sin\left(\frac{\pi x}{l}\right) \tag{5.2.29}$$

Note that $\gamma(x)$ will be used from now on to indicate an eigenfunction. This distinguishes it from the more general $\mathbf{x}(x)$, which is used to indicate the general spatial part of a solution before the particular boundary conditions have been applied. Also, when it's being used in a complicated equation, the (x) will be dropped to keep the expression clear.

This mode is shown in Figure 5.8. Note that it indicates zero deflection at the bar ends and a maximal deflection in the middle of the bar. The overall time-varying behavior of this mode is given by

$$\xi(x,t) = a_1 \sin\left(\frac{\pi x}{l}\right)\cos(\omega_1 t + \phi_1) \tag{5.2.30}$$

where $\omega_1 = \sqrt{\frac{EA}{m}}\frac{\pi}{l}$. The constants a_1 and ϕ_1 are necessary to match the particular initial conditions of the problem.

Figure 5.9 shows what the bar deformations would be like over one oscillation cycle. The period of the oscillation is T ($T = \frac{2\pi}{\omega_1}$). For simplicity, a_1 is taken to be 1.0 and ϕ_1 is set to zero. It is clear from this plot that the deflected shape is always in the form of a half sine wave—only the magnitude changes. As time progresses from zero, the deflected positions reduce in magnitude, going to zero at

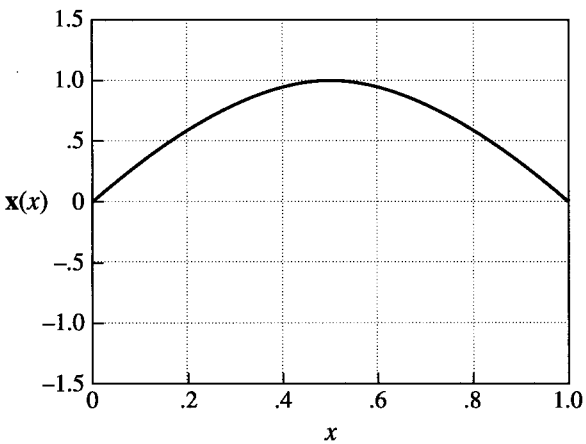


Figure 5.8 First mode of a fixed-fixed bar

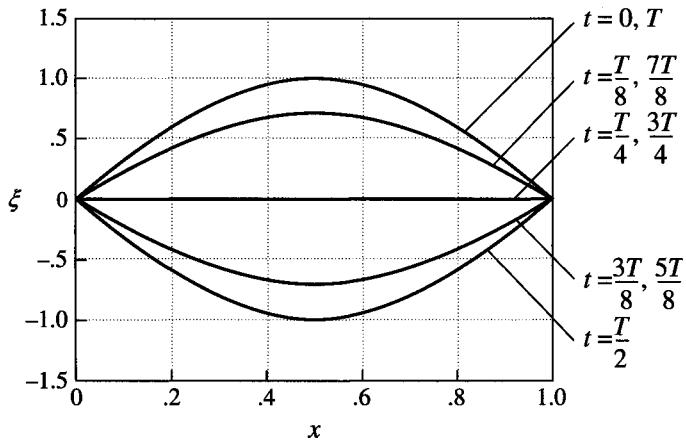


Figure 5.9 Time behavior of fixed-fixed bar's first mode

$t = \frac{\pi}{2\omega_1}$, reaching a maximum negative deflection at $t = \frac{\pi}{\omega_1}$, and returning to the original deflections at $t = \frac{2\pi}{\omega_1}$.

This is an exact analogue of the discrete eigenvector behavior first seen in Chapter 4. If the bar is deflected into the first mode and then released, it will continue to oscillate with the same continuous modal deflection, and at a frequency equal to ω_1, just as discrete systems can oscillate in particular modal vectors. The difference between this continuous system and those of Chapter 4 is that in this case we have an infinite number of mode shapes. And this makes sense. A continuous system, such as the bar we're looking at, can be thought of as a collection of an infinite number of infinitesimal mass and spring elements. Since the number of mode shapes depends upon the number of masses in a discrete system, we should expect an infinite number of modes when we've got an infinite number of masses. In fact, part of the motivation behind the approximate methods (discussed in Chapter 6) is to try to approximate the infinite number of mass elements of a continuous structure by a finite number of masses.

The remaining mode shapes for our bar are simply sines with more and more included segments (Figure 5.10). As the mode shape becomes more "wiggly," the frequency of oscillation also rises. This is a characteristic of vibrating systems—more "wiggles" in the mode shape implies a higher frequency of oscillation.

Now that we've gotten all our mode shapes pinned down, we can determine the system's response to a set of initial conditions. The general solution is given by a summation of all the possible modal responses:

$$\xi(x,t) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{l}\right) \cos(\omega_n t + \phi_n) \tag{5.2.31}$$

The particular values of the a_n and ϕ_n will depend on the particular initial conditions. In general, we'll have a some initial displacement, $f(x)$, and velocity, $g(x)$, for all points of the bar. Thus,

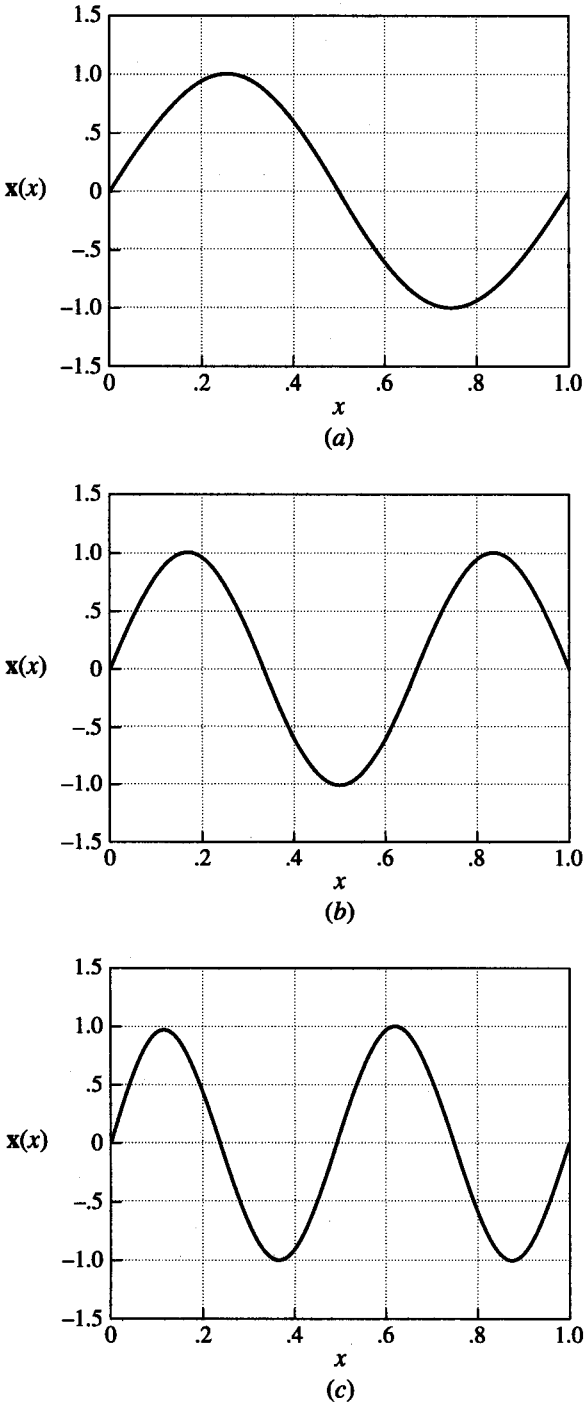


Figure 5.10 Second, third, and fourth modes of a fixed-fixed bar

evaluating (5.2.31) at $t = 0$ will give us the two equations

$$f(x) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{l}\right) \cos(\phi_n) \tag{5.2.32}$$

and

$$g(x) = -\sum_{n=1}^{\infty} a_n \omega_n \sin\left(\frac{n\pi x}{l}\right) \sin(\phi_n) \tag{5.2.33}$$

We'll further explore the meaning of these summations by looking at a particular set of initial conditions. The most immediately understandable initial conditions are those in which the system under consideration has been displaced into some initial configuration and then released. In this case, all the ϕ_n's will equal zero. This can be seen from an examination of (5.2.33). If all the velocity of all points along the bar are zero at $t = 0$, then having all ϕ_n's equal zero makes $g(x)$ zero, since all the individual terms in the summation will be zero. We're then left with

$$f(x) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{l}\right) \tag{5.2.34}$$

Note that the $\cos(\phi_n)$ is 1.0 since all the ϕ_n's are zero.

If we look at (5.2.34), we see that it is *exactly* in the form of a Fourier series! The $\sin(n\pi x)$ terms are precisely the harmonics we would use if we were trying to construct a Fourier series to approximate a physical displacement that had to start at zero (at $x = 0$) and end at zero (at $x = l$). The a_n are the unknown Fourier coefficients. And, since we now know how to deal with Fourier series problems, we are fully capable of determining the a_n for this problem.

Although this example was a bit easier to study than the most general problem, there are no real difficulties with the fully general problem (for which both $f(x)$ and $g(x)$ are nonzero), other than an increased workload.

Let's be clear now about what we've got here. A quick glance at Appendix A (come on, a quick glance won't kill you) shows us that as advertised at the start of the chapter, we now know the free responses for a circular rod, a tensioned string, and a one-dimensional acoustic duct. So let's say you were interested in building a guitar. You can now analytically predict the necessary length, materials, and tension needed for the strings to get the particular pitch you're after. You can also evaluate the differences in pitch that will exist between different string materials. Thus you have a guide to whether you'd want to use nylon or steel for your strings. Or let's say you've decided to go into the pipe organ business. You can now figure out how long your pipes should be for them to give the right notes. Or maybe you want to determine the value of EA experimentally. If so, you can set up a bar experiment like the one described here and use your transducers to observe the response of the system to an initial excitation, record the frequencies of the response, and by correlating these with your theoretical results (which involve EA), deduce what EA must be for your test specimen.

Bottom line—this stuff isn't trivial by any means, and you're in a good position to do quite a bit of engineering just with the results we've gotten so far. However, our life won't be complete until we've looked at a beam. So, on to the next section we go (after an example or two).

Example 5.1

Problem Find the eigenfunctions and natural frequencies of the spring restrained, tensioned string shown in Figure 5.11. $\rho = .002$ kg/m, $T = 21$ N/m, $l = 1$ m, $k = 25$ N/m.

Solution From Appendix A we find that the equation of motion for a uniform, tensioned string is given by

$$Ty_{xx} = \rho\ddot{y}$$

Comparing this to (5.2.8) shows that the correspondence between the rod equation and the string equation is

$$\xi \leftrightarrow y$$
$$m \leftrightarrow \rho$$
$$EA \leftrightarrow T$$

Thus we can immediately write the general solution as

$$\mathbf{x}(x) = b_1 \cos(\beta x) + b_2 \sin(\beta x)$$

where $\beta \equiv \omega\sqrt{\frac{\rho}{T}}$.

The boundary condition at $x = l$ is $y(l, t) = 0$. At $x = 0$ we need to have a force balance between the deflected spring and the tensioned string. If we sketch the attachment point A in a deflected position, these forces can be drawn as shown in Figure 5.12.

For small angles of θ the approximations

$$\sin(\theta) \approx \theta \approx \frac{dy}{dx}$$

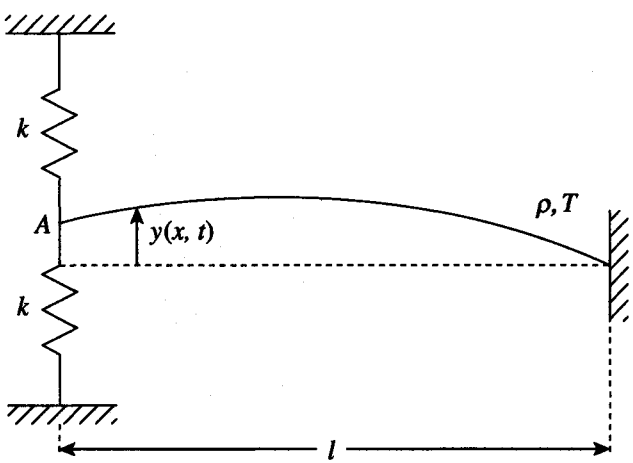


Figure 5.11

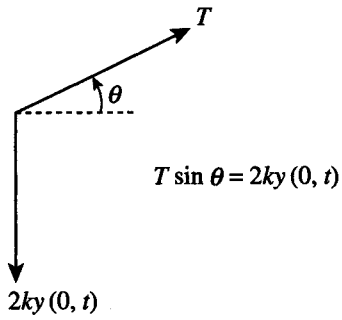


Figure 5.12

hold. Using this gives us the boundary condition

$$T\frac{dy(0,t)}{dx} = 2ky(0,t)$$

as our left-end boundary condition. Using

$$\mathbf{x}(x) = b_1\cos(\beta x) + b_2\sin(\beta x)$$

and

$$\mathbf{x}_x(x) = \beta\left(-b_1\sin(\beta x) + b_2\cos(\beta x)\right)$$

and evaluating at the two end points gives us the new boundary conditions

$$\beta T b_2 = 2kb_1 \tag{5.2.35}$$

$$b_1\cos(\beta l) + b_2\sin(\beta l) = 0 \tag{5.2.36}$$

Equation (5.2.35) implies that $b_2 = \frac{2k}{\beta T}b_1$ which, when substituted into (5.2.36), leads to

$$\tan(\beta l) = -\frac{T\beta l}{2kl}$$

Graphically this looks like the plot in Figure 5.13.

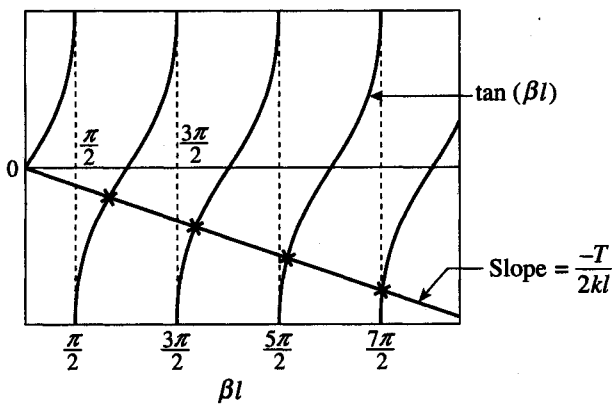


Figure 5.13

As you can see, reducing $\frac{T}{2kl}$ toward zero (infinitely stiff springs) moves the $\beta_i l$ roots toward multiples of π. This corresponds to a fixed-fixed solution, analogous to the bar example worked in the beginning of the chapter.

For the given values of T, k, ρ, and l, we have to solve

$$\tan(\beta) = -.42\beta$$

the first two solutions of which are $\beta_1 = 2.3605$ and $\beta_2 = 5.1458$.

These correspond to natural frequencies of

$$\omega_1 = \beta_1\sqrt{\frac{T}{\rho}} = 2.3605\sqrt{\frac{.21}{.002}} = 241.9\,\text{rad/s}$$

$$\omega_2 = \beta_2\sqrt{\frac{T}{\rho}} = 5.1458\sqrt{\frac{.21}{.002}} = 527.3\,\text{rad/s}$$

Since

$$\begin{aligned}\mathbf{x}(x) &= b_1\cos(\beta x) + \left(\frac{2k}{\beta T}\right) b_1 \sin(\beta x)\\ &= b_1\left(\cos(\beta x) + \frac{2k}{\beta T}\sin(\beta x)\right)\end{aligned} \tag{5.2.37}$$

we'll finally obtain (ignoring the constant b_1)

$$\mathbf{x}_1(x) = \cos(2.3605x) + 1.0087\sin(2.3605x)$$
$$\mathbf{x}_2(x) = \cos(5.1458x) + .4627\sin(5.1458x)$$

as the first two eigenfunctions.

Example 5.2

Problem Determine the β equation for a uniform string that's fixed on one end and has both a spring and a mass attached to the other (only vertical motions allowed), as in Figure 5.14

Solution The left boundary condition is given by $y(0, t) = 0$. The right boundary condition is found from a force balance, as in Figure 5.15.

Summing forces in the vertical direction gives

$$m\ddot{y}(l, t) = -ky(l, t) - T\sin(\theta)$$

Since $\theta = \tan^{-1}\left(\frac{dy}{dx}\right)$ and $\frac{dy}{dx}$ is small, we have

$$\ddot{y}(l, t) = -ky(l, t) - T\frac{dy(l, t)}{dx}$$

From Example 5.1 we know that

$$\mathbf{x}(x) = b_1\cos(\beta x) + b_2\sin(\beta x)$$

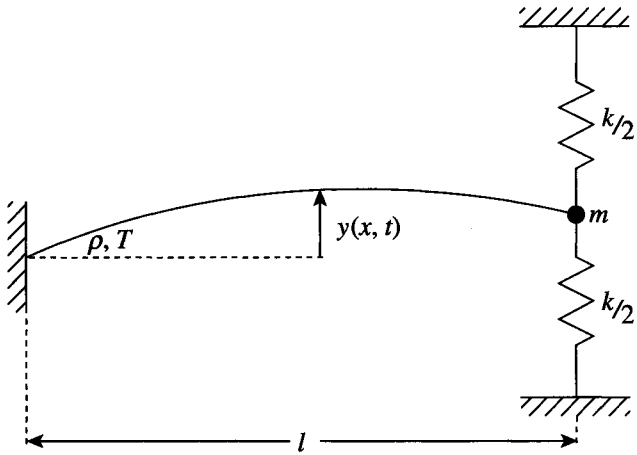


Figure 5.14

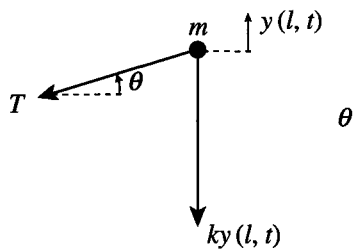


Figure 5.15

where $\beta \equiv \omega\sqrt{\frac{\rho}{T}}$. Using $\mathbf{x}(0) = 0$ yields $b_1 = 0$. Applying the right boundary condition gives

$$-m\omega^2\mathbf{x}\Big|_{x=l} = -k\mathbf{x}\Big|_{x=l} - T\frac{dy}{dx}\Big|_{x=l}$$

$$-m\omega^2 \sin(\beta l) = -k\sin(\beta l) - T\beta\cos(\beta l)$$

Using $\omega^2 = \beta^2\frac{T}{\rho}$ and simplifying, we have

$$\tan(\beta l) = \frac{-T\beta}{k - \frac{T}{\rho}\beta^2}$$

5.3 FREE VIBRATION OF A BEAM

In Section 5.2, we saw how to analyze the free vibration of a uniform bar. If the bar materials are known exactly and are completely uniform, then our results are theoretically exact also. Now we'll look at a more complicated question, namely, how the transverse vibrations of a beam behave.

Almost everyone reading this book has already had a course in deformable bodies (that's the good news), and almost everyone has forgotten just what was covered in the course (that's the bad news). If you're one of the fortunate few who can recall the governing equations of a beam, congratulations.

You can skip a few pages. But for the rest of you, follow along for a bit as we rediscover the wonderful world of beams.

A system we know how to handle that supports transverse vibrations (as the beam does) is a tensioned string. From Appendix A we see that the governing equations are just like those of a bar, something we're familiar with. Basically, the difference between a beam and a string is that a string is limp and floppy unless it's under tension, whereas a beam is stiff even without any tensile load. It's this additional characteristic, an internal bending stiffness, that makes the beam equations more complex. We'll derive the complete equation of motion, including applied forces here, but we won't deal with the effect of forcing immediately.

Figure 5.16 shows what our beam looks like and also shows an infinitesimal element taken out of the beam. We shall be deriving the simplest beam model, namely the Bernoulli-Euler beam model, in this section. There are two basic simplifications involved in this model, and the first has to do with the shape of the deformed element. You'll note that the beam element is drawn as a parallelogram. Actually, the drawing would be more accurate if we drew this piece as a curved element. In this way we'd be including the shear deformation of the beam. Thus, we've reached our first approximation. We're going to assume that the deformations due to bending are much larger than those due to shear, and so we'll be neglecting shear deformation effects. A more complete model, called a Timoshenko beam model, includes this additional effect. As long as the beam doesn't "wiggle" too much, this approximation is a good one. That is, it will start to break down as we go to higher and higher modes of vibration.

The next step in determining the equations of motion for the beam is to simply apply to the beam element what you've learned in your dynamics class. We're allowing only vertical motions of the beam, so we'll derive the equation of motion for the vertical direction only. Also, the element could conceivably rotate—which would require us to use one of the several formulas that deal with moments

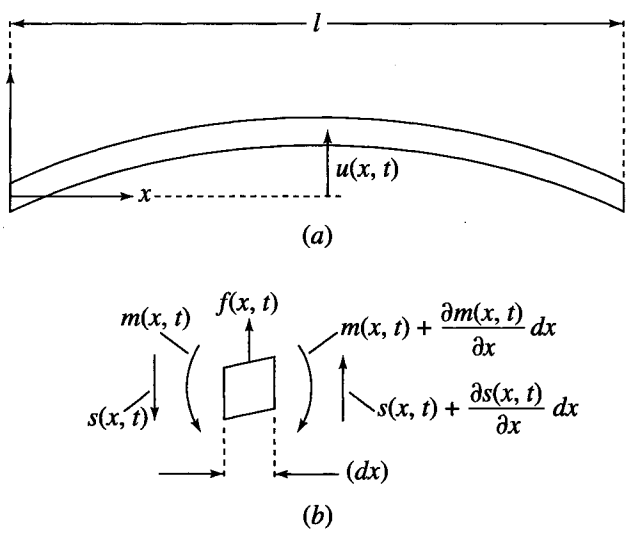


Figure 5.16 Differential analyses of a beam

and rotation. At this point we'll invoke our second assumption. We'll assume that the rotational inertia of our beam element is negligibly small compared with the translational inertia. As long as our beam is much longer than it is high, this will be a reasonable approximation. However, as the beam becomes "stubbier," we'll start to see error creep in because of our assumption. Once again, Timoshenko beam theory will include this effect. As a rule of thumb, as long as the length is about 10 times greater than the height, neglecting rotational inertia will be a reasonable assumption.

Applying a force balance to the differential element dx of Figure 5.16 will give us

$$f(x,t)dx + \left(s(x) + \frac{\partial s(x)}{\partial x}dx\right) - s(x) = \rho(x)\ddot{u}\,dx \tag{5.3.1}$$

where $s(x)$ represents the shear force within the beam and $\rho(x)$ is the mass per unit length along the beam. Canceling out $s(x)$ and dividing through by the common factor of dx leaves us with

$$f(x,t) + \frac{\partial s(x)}{\partial x} = \rho(x)\ddot{u} \tag{5.3.2}$$

Next, let's look at the rotational part of the problem. Summing moments (in the z direction) about the center of mass of our differential element will give us

$$m(x) + \frac{\partial m(x)}{\partial x}dx - m(x) + \left(s(x) + \frac{\partial s(x)}{\partial x}dx\right)\frac{dx}{2} + s(x)\frac{dx}{2} = 0 \tag{5.3.3}$$

where $m(x)$ is the moment acting on the element. There's no inertial term on the right because we're ignoring rotational inertia by assumption. Canceling out the m's, dividing by dx, and taking the limit of dx going to zero then gives us

$$\frac{\partial m(x)}{\partial x} = -s(x) \tag{5.3.4}$$

Using (5.3.4) in (5.3.2) yields

$$\rho(x)\ddot{u} = f(x,t) - \frac{\partial^2 m(x)}{\partial x^2} \tag{5.3.5}$$

Now we need only one more piece of information, namely, the relationship between m and u. Flipping back through whatever text you used in your deformable bodies course, you'll find that this relationship is

$$m(x) = EI(x)\frac{\partial^2 u(x)}{\partial x^2} \tag{5.3.6}$$

where $EI(x)$ is the bending stiffness of the beam. Combining (5.3.6) and (5.3.5) gives us our final beam equation:

$$\rho(x)\ddot{u} = -\left(EI(x)u_{xx}\right)_{xx} + f(x,t) \tag{5.3.7}$$

The difference between this equation and the one governing the bar's behavior is that instead of two partial differentiations with respect to x, the beam equation has four. If you ignore this detail, you'll

see that it's simply an equation that equates an inertial force (in this case $\rho\ddot{u}$) with the forces due to the internal workings of the beam (those involving EI) and any externally applied loads $f(x,t)$.

Now that we've got the equation of motion, we can set $f(x,t)$ to zero and solve for the free vibration of the beam. Just as we did for the bar problem, we're going to use separation of variables, i.e., we'll assume that the part of the motion depending on time can be separated out from that part depending on position. If we let $u(x,t) = \mathbf{x}(x)\mathbf{t}(t)$ and divide by $\rho\mathbf{xt}$ (5.3.7) becomes

$$\frac{\ddot{\mathbf{t}}}{\mathbf{t}} = -\frac{(EI(x)\mathbf{x}_{xx})_{xx}}{\rho(x)\mathbf{x}} \tag{5.3.8}$$

(Note that all the spatially dependent components are now grouped together.)

Just as we did in the bar example, we'll assume that the physical properties of the beam are constant over the beam. Thus (5.3.8) simplifies to

$$\frac{\ddot{\mathbf{t}}}{\mathbf{t}} = -\frac{EI\mathbf{x}_{xxxx}}{\rho\mathbf{x}} \tag{5.3.9}$$

Once again, a time-dependent quantity can equal a spatially varying quantity on if they're both equal to the same constant. Since we now expect the time-dependent behavior to be in the form of an oscillator, we can immediately set the constant equal to $-\omega^2$, giving us the two equations

$$\frac{\ddot{\mathbf{t}}}{\mathbf{t}} = -\omega^2 \tag{5.3.10}$$

and

$$-\frac{EI\mathbf{x}_{xxxx}}{\rho\mathbf{x}} = -\omega^2 \tag{5.3.11}$$

Equation (5.3.10) can be rearranged to give us the same oscillator equation we found with the bar example,

$$\ddot{\mathbf{t}} + \omega^2\mathbf{t} = 0 \tag{5.3.12}$$

The new discovery is the equation we obtain from (5.3.11),

$$\frac{EI}{\rho}\mathbf{x}_{xxxx} - \omega^2\mathbf{x} = 0 \tag{5.3.13}$$

If we rearrange this a bit, we can get it into a form that resembles an oscillator equation

$$\mathbf{x}_{xxxx} - \frac{\omega^2\rho}{EI}\mathbf{x} = 0 \tag{5.3.14}$$

except that we have four differentiations with respect to x instead of two and we've got a minus sign instead of a plus on the second term.

If we're to get any further, we're going to have to figure out what the solutions to (5.3.14) might be. Before we do this, let's clean up the notation just a bit. Rather than writing $\frac{\omega^2\rho}{EI}$ all the time, we'll

replace it with β^4. You'll recall we did something similar in the bar example. Thus we're left with

$$\mathbf{x}_{xxxx} - \beta^4 \mathbf{x} = 0 \tag{5.3.15}$$

The solutions have to be functions that, when differentiated four times, give us back the original function (multiplied by some constant). This is because we have to subtract $\beta^4\mathbf{x}$ from it, which is simply the original function times β^4. Luckily, there are four well-known functions that operate this way: sin, cos, sinh, and cosh! So even though the equation looks more complicated that a simple oscillator, it has the exact same solutions (sin and cos) and also gives us two new solutions (sinh and cosh). More specifically, our general solution is

$$\mathbf{x}(x) = b_1 \cos(\beta x) + b_2 \sin(\beta x) + b_3 \cosh(\beta x) + b_4 \sinh(\beta x) \tag{5.3.16}$$

You'll note that this solution, when differentiated four times, is exactly cancelled out by β^4 times itself, thus satisfying (5.3.15). For those who don't recall what sinh and cosh are, they're simply given by

$$\sinh(x) = \frac{e^x - e^{-x}}{2} \tag{5.3.17}$$

and

$$\cosh(x) = \frac{e^x + e^{-x}}{2} \tag{5.3.18}$$

Thus both functions get very large for large values of x. Only for small x is there much difference between them, with $\sinh(x)$ going to 0 as x goes to zero and $\cosh(x)$ going to 1.

What we have here is simply a richer choice of possible mode shapes than we had with the bar problem. The sin and cos terms give us the "wiggle" in our solutions, while the sinh and cosh terms give us solutions that get large quickly as x increases. By putting these components together, we can find the solutions for beam vibrations having any possible set of boundary conditions. And now, having mentioned boundary conditions, we should probably move on and examine this subject.

Because the beam has bending stiffness, its boundary conditions are more complicated than those found in the bar example. In that case, only one boundary condition was needed at each end of the bar to fully describe the physical setup of the problem. But now we're going to need *two* boundary conditions at each end. Figure 5.17 shows some of the possible boundary conditions commonly associated with beam vibration problems. Figure 5.17*a* represents a clamped (or cantilevered) boundary condition. For this one we visualize the beam as being immovable, locked into a wall. Thus the appropriate boundary conditions are $u(0,t) = \frac{\partial u(0,t)}{\partial x} = 0$; i.e., the displacement and slope at the clamped end are zero. The next possibility (Figure 5.17*b*) is a pinned boundary condition. In this case we're constraining the displacement to be zero ($u(0,t) = 0$), but we allow finite slopes. What *aren't* allowed are moments. This is one of the results that should have shown up in your deformable bodies course or perhaps in a dynamics course. If you didn't see

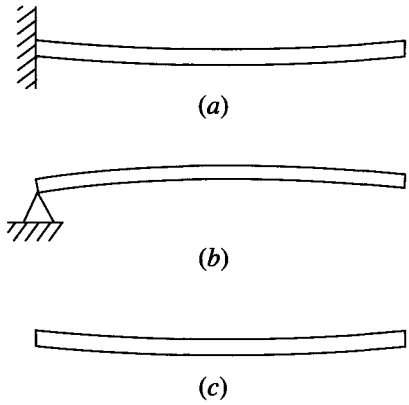


Figure 5.17 (*a*) Clamped boundary condition; (*b*) pinned boundary condition; (*c*) free boundary condition

it there, you know it now—pinned joints imply zero moments. So $\frac{\partial^2 u(0,t)}{\partial x^2} = 0$. The final example (Figure 5.17*c*) is that of the free boundary condition. In this case we're not constraining the displacement or slope—both are in general nonzero. But the moment and shear at this point must equal zero, i.e.,

$$\frac{\partial^2 u(0,t)}{\partial x^2} = 0 \quad \text{and} \quad \frac{\partial^3 u(0,t)}{\partial x^3} = 0$$

Taking a moment to look at both the string problem and the beam problem reveals something important. For the string problem, the dependent variable is differentiated twice with respect to x (y_{xx}). Also, the problem has two boundary conditions, with one applied at each end. Two differentiations and two boundary conditions. In the beam problem we see that the greatest number of differentiations is four (u_{xxxx}) and the problem has four boundary conditions, with two at each end. Four differentiations and four boundary conditions. There seems to be a pattern here, namely, that the number of boundary conditions equals the highest number of differentiations of the dependent variable. Since half the boundary conditions are applied to one end and half to the other, you'll often see people characterize a continuous problem as being of order $2p$, where $2p$ is the highest number of differentiations and p is therefore the number of boundary conditions you'll need to apply at each end. (Keep in mind that we're talking about one-dimensional problems here; the same general concept holds for plates and other higher dimensional problems.)

We should also take this time to give names to the different kinds of boundary condition we've just encountered. When a boundary condition involves the geometric configuration of the structure, like the displacement or the slope at an end point, we call it a *geometric boundary condition*. The other type of boundary condition involves the shear force or the moment at an end point, a so-called *natural boundary condition*. A pinned end of a beam therefore involves both types: the geometric boundary condition is the one that says $u = 0$, and the natural boundary condition is the one that says $u_{xx} = 0$. The distinction between boundary conditions will be important when we look at approximation techniques in Chapter 6.

Example 5.3

Problem In this example we'd like to determine the response of a free-free beam, i.e., one for which both the left and right end are unrestrained. Thus this might be considered to be the simplest model for the behavior of an airplane or satellite.

Solution We must keep in mind that this is an unrestrained support condition (like floating in space), and so we'll have rigid body motion as well as vibrations. The analysis to follow is concerned solely with the vibrations due to the internal stiffness of the beam. Any rigid body modes can be solved for separately if needed, a problem we'll consider after we've looked at the flexible modes of the system. For transverse vibrations of the sort we're looking at, they'll correspond to rigid translation in the u direction and rigid rotation.

All we need do now is apply the boundary conditions

$$\frac{\partial^2 u(0,t)}{\partial x^2} = \frac{\partial^2 u(l,t)}{\partial x^2} = \frac{\partial^3 u(0,t)}{\partial x^3} = \frac{\partial^3 u(l,t)}{\partial x^3} = 0$$

to (5.3.16). Differentiating (5.3.16) the appropriate number of times, applying the boundary conditions, and using the fact that $\cos(0) = 1$, $\sin(0) = 0$, $\cosh(0) = 1$, and $\sinh(0) = 0$ gives us

$$\beta^2(-b_1 + b_3) = 0$$

$$\beta^3(-b_2 + b_4) = 0$$

$$\beta^2\left(-b_1\cos(\beta l) - b_2\sin(\beta l) + b_3\cosh(\beta l) + b_4\sinh(\beta l)\right) = 0$$

and

$$\beta^3\left(b_1\sin(\beta l) - b_2\cos(\beta l) + b_3\sinh(\beta l) + b_4\cosh(\beta l)\right) = 0$$

The first two equations tell us that $b_1 = b_3$ and $b_2 = b_4$ (since letting $\beta = 0$ is clearly an uninteresting solution). Using this bit of knowledge, and factoring out the β terms, lets us rewrite the third and fourth equations as

$$\begin{aligned} b_1\left(\cosh(\beta l) - \cos(\beta l)\right) + b_2\left(\sinh(\beta l) - \sin(\beta l)\right) &= 0 \\ b_1\left(\sin(\beta l) + \sinh(\beta l)\right) + b_2\left(\cosh(\beta l) - \cos(\beta l)\right) &= 0 \end{aligned} \tag{5.3.19}$$

or

$$\begin{bmatrix} (\cosh(\beta l) - \cos(\beta l)) & (\sinh(\beta l) - \sin(\beta l)) \\ (\sin(\beta l) + \sinh(\beta l)) & (\cosh(\beta l) - \cos(\beta l)) \end{bmatrix} \begin{Bmatrix} b_1 \\ b_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

You'll note that, just when it seemed we'd left the exciting field of finite-dimensional eigenvalue problems, they're back, since we've got something in the form $[A]X = 0$. And from what we've already learned, we know that there'll be a solution only if $|[A]| = 0$. Taking this determinant gives us

$$(\cosh(\beta l) - \cos(\beta l))(\cosh(\beta l) - \cos(\beta l)) - (\sin(\beta l) + \sinh(\beta l))(\sinh(\beta l) - \sin(\beta l)) = 0$$

Expanding this out and using the trigonometric identities $\sin^2 + \cos^2 = 1$, along with $\cosh^2 - \sinh^2 = 1$, we'll finally obtain

$$\cos(\beta L)\cosh(\beta L) = 1$$

Thus, the particular values of β, those that will determine the mode shapes and associated oscillation frequencies from our beam, are found from this single equation. We'll denote these particular solutions as β_n. Figure 5.18 plots both $\cos(\beta l)\cosh(\beta l)$ and 1. These two curves intersect at $\cos(\beta l)\cosh(\beta l) = 1$, indicating the associated values of $\beta_n l$.

You can see from Figure 5.18 that the β values no longer increase in a regular manner, as they did for the bar problem (where $\beta_n = \frac{n\pi}{l}$). The first three nonzero values of $\beta_n l$ are 4.7300, 7.8532, and 10.9956. Although these don't differ from each other by a constant amount, the difference *approaches a constant* as we go to higher and higher modes. This is actually predictable if we think about what's happening with the cos and cosh terms. The cosh is getting big very quickly as the mode number increases, and thus cos has to be very, very small for their product to equal 1. Therefore the roots approach those β's at which $\cos(\beta l)$ is zero, namely, $\frac{(2n-1)\pi}{2}$. It's only when the cosh isn't really huge (as in the lower modes) that the roots move substantially away from the $\cos(\beta l) = 0$ positions.

Now we can find the exact mode shape associated with each of the $\beta_n l$ values.

Using (5.3.19) to solve for b_1 in terms of b_2 gives us

$$b_2 = b_1 \frac{\sin(\beta_n l) + \sinh(\beta_n l)}{\cos(\beta_n l) - \cosh(\beta_n l)} \tag{5.3.20}$$

Since $b_1 = b_3$ and $b_2 = b_4$, (5.3.16) becomes

$$\mathbf{x}(x) = b_1 \left(\cos(\beta_n x) + \cosh(\beta_n x)\right) + b_2 \left(\sin(\beta_n x) + \sinh(\beta_n x)\right)$$

Making use of (5.3.20) then gives us

$$\gamma(x) = b_1 \left(\cos(\beta_n x) + \cosh(\beta_n x) + \frac{\sin(\beta_n l) + \sinh(\beta_n l)}{\cos(\beta_n l) - \cosh(\beta_n l)} \left(\sin(\beta_n x) + \sinh(\beta_n x)\right) \right) \tag{5.3.21}$$

Now (5.3.21) looks pretty complicated because it is complicated. But the important thing to remember is that it is fundamentally no different from the simple sine and cosine solutions we found for the bar problem. It's simply the mathematical representation of the actual mode shapes for our problem. It seems so complicated only because we're not used to dealing with combinations of sines and

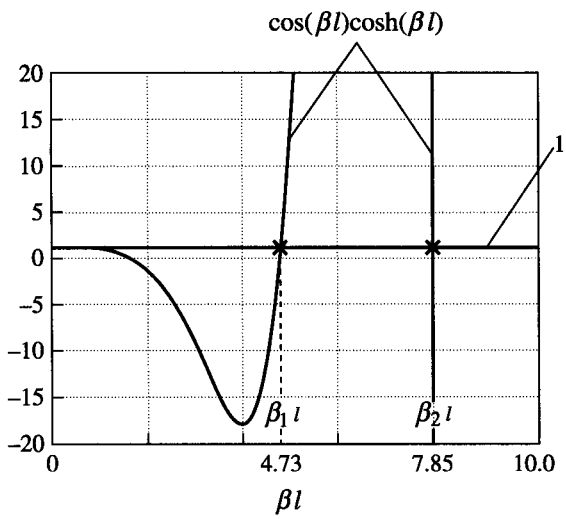


Figure 5.18 $\beta_n l$ values for a free-free beam

hyperbolic sines and such. And if you look at Figure 5.19, you'll see that the system's modes look like quite reasonable mode shapes, just the sort you might expect in a freely vibrating beam. (Keep in mind that the plots are massively exaggerated in scale; the actual vibration amplitudes would normally be too small to be visible to the naked eye.)

Even though the equations are complex, it doesn't really matter in a practical sense unless you're planning on evaluating them by hand. But if you ever plan to really manipulate these modes, you'll undoubtedly use a computer (which won't be at all upset because there's more going on here than just sines and cosines).

Actually, now that we're on the subject, it's a good idea to consider just what would happen if (5.3.21) were entered into a computer program. The reason for wanting to look at this is that there are pitfalls associated with beam solutions, pitfalls that have caused hours of frustration to many students who didn't think carefully enough about potential problems before running their programs.

The main problem lies in the behavior of cosh and sinh. Recall two things:

$$\cosh(x) = \frac{e^x + e^{-x}}{2} \quad \text{and} \quad \sinh(x) = \frac{e^x - e^{-x}}{2}$$

Both these functions grow very large, very quickly. Since the argument of our mode shape depends on βx, it's clear that for large β's, we'll have large cosh and sinh terms. Furthermore, as their arguments grow, the difference between a cosh and a sinh goes to zero. A glance at the exponential components of $\sinh(\beta x)$ and $\cosh(\beta x)$ shows that the difference between them is simply $e^{-\beta x}$, something that goes to zero very quickly as βx increases.

Look carefully at the mode shape given by (5.3.21). If we make the approximation that $\sin(\beta_n l) + \sinh(\beta_n l)$ is basically equal to $\sinh(\beta_n l)$ for large values of β_n and $\cos(\beta l) - \cosh(\beta l)$ is basically equal to $-\cosh(\beta_n l)$, then their ratio is just about equal to 1. So we're left with $\mathbf{x} = b_1[\cos(\beta x) + \cosh(\beta x) - (\sin(\beta x) + \sinh(\beta x))]$. Since $\sinh(\beta_n x)$ and $\cosh(\beta_n x)$ are basically equal for large β_n, $\mathbf{x}$ is approximately equal to $b_1[\cos(\beta_n x) - \sin(\beta_n x)]$. Although this isn't actually a good enough approximation, it is sufficient to illustrate some points. The first thing to realize is that a computer is going to have a much easier time computing $\cos(\beta_n x) - \sin(\beta_n x)$ than it will with the more exact expression. Thus we'd save computation time by using the approximate solution. Second, the computer will actually be unable to calculate the exact expression if too high a mode is being considered. For instance, $\cosh(30)$ is roughly 5.34×10^{12}. When you're dealing with numbers this large, even the best computers will start to have difficulties. Why? Well, consider that you're asking the computer to compute $\cos(\beta_n l) - \cosh(\beta_n l)$. If $\beta_n l$ is 30, then you're asking the computer to accurately add together -5.34×10^{12} plus something that's around 1. There's no way you're going to do that and still retain accuracy. And take the ratio itself (which we've been saying is around 1). Well, we have to be pretty darn sure what it's precisely equal to, since we're multiplying by $\sinh(\beta_n x)$, which is going to be huge for values of x away from 0.

Finally, we've got to be concerned that some of our arguments involve $\beta_n l$ while others involve $\beta_n x$. Since x goes from 0 to l, it's clear that $\cosh(\beta_n x)$ will vary between zero and very large values. Thus it's not true everywhere on the beam that $\cosh(\beta_n x)$ is approximately equal to $\sinh(\beta_n x)$ and both are much greater than $\cos(\beta_n x)$ and $\sin(\beta_n x)$; it's true only well away from $x = 0$.

What we have to do in a case like this is carefully construct an approximation to the actual mode shape that takes everything we've been talking about into account. Luckily for us, most computers will be fine for the first half-dozen modes or so. Thus it isn't necessary to get into this any further unless you're really interested in getting an excellent grasp of modal analysis or are doing detailed analyses. If you have an interest in reading further, Reference 2 goes into more detail on how to construct appropriate approximations.

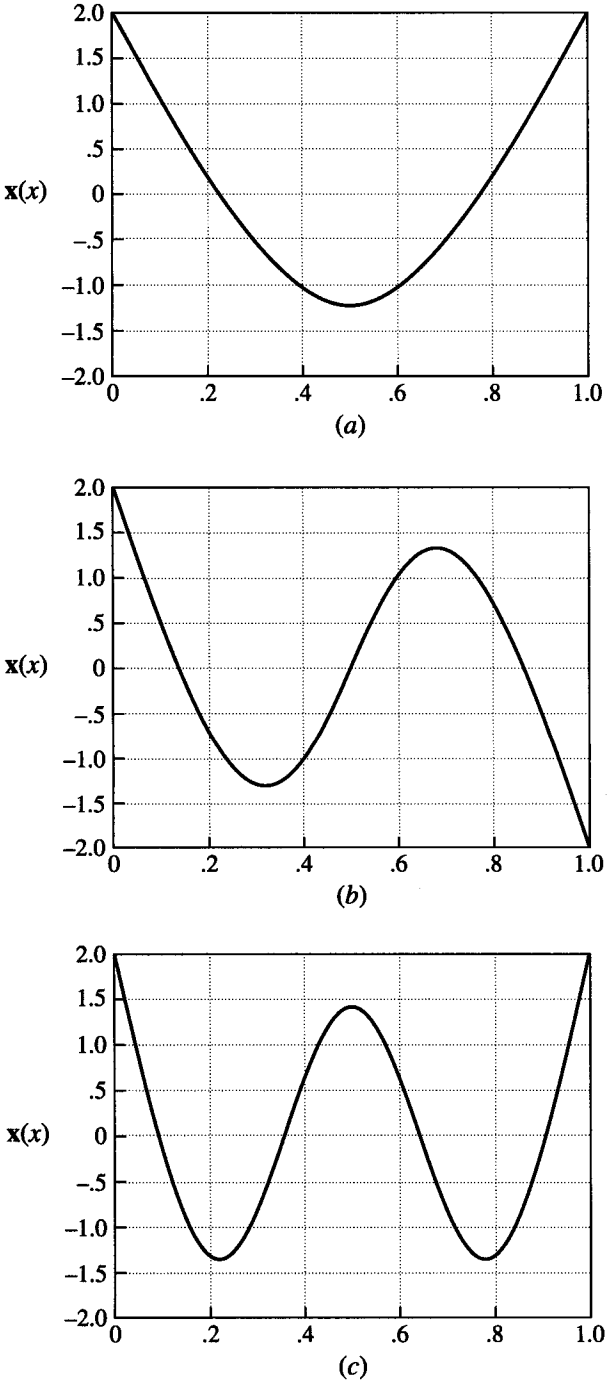


Figure 5.19 (*a*) First mode of a free-free beam; (*b*) second mode of a free-free beam; and (*c*) third mode of a free-free beam

Before leaving this section, it's time to make a mention of rigid body modes. All the elastic modes we've looked at so far were associated with nonzero values of βl. However, we can see from Figure 5.18 that $\beta l = 0$ is also a valid solution. To see what this implies in terms of modal responses, we need to go back to (5.3.15). If $\beta l = 0$ and l isn't zero, then the unforced equation governing the spatial deflections becomes

$$\mathbf{x}_{xxxx} = 0 \tag{5.3.22}$$

This is straightforward to integrate, and doing so gives us

$$\mathbf{x}(x) = c_1 x^3 + c_2 x^2 + c_3 x + c_4 \tag{5.3.23}$$

The boundary conditions $\mathbf{x}_{xx} = \mathbf{x}_{xxx} = 0$ at $x = 0$ mean that $c_1 = c_2 = 0$. This leaves us with $\mathbf{x} = c_3 x + c_4$. This also satisfies the boundary conditions at $x = l$. Since we've got two undetermined constants, we can define two mode shapes. A reasonable choice is to break up the possible motion into a rigid body translational mode and rigid body rotational mode. Thus, we can set the first mode equal to a constant d_1 and the second equal to $d_2(x - \frac{l}{2})$. In this way the first mode is symmetric and the second is antisymmetric with respect to the center of the beam. These modes are quite important when one is actually analyzing freely supported systems, such as airliners or spacecraft, since they represent motion that occurs at zero frequency. The zero frequency implies that, once the object is pushed into one of these modes, it will continue moving in that way. Thus, if a satellite is given an initial translational velocity, it will keep translating. Since this is often undesirable, we have to keep track of these motions (through our rigid body modes) so we can be sure to leave them unexcited and/or control them.

With the knowledge you now have, you can address a wide range of problems. For instance, imagine that you're working for a tennis racket manufacturer and you want to improve your racket line. Your first dynamics course may have covered how you can find the center of percussion, i.e., the point of a rigid bar at which impact forces will not impart reaction forces to a supporting pivot (which in this case would be your hand). But this analysis completely ignores the flexibility of tennis rackets. You'll want to consider what the modes are for a tennis racket. If you started your analysis by assuming that the racket is well modeled by a beam, you could use free–free boundary conditions (on the assumption that the inertial forces of the ball-racket interaction outweigh the forces of your hand). Then you could go through an analysis similar to the one we've just done. Finally, knowing the modes of the racket, you could look at them and determine where they're the greatest and where they're equal to zero. Presumably, you'd like the point at which you grasp the racket to be where a node (point of zero deflection) is located, since this probably would be the most comfortable for the player. Of course, if you wanted to be really accurate, you'd use something more involved than a uniform beam model, but you'd get an easy first estimate from using the simple beam approximation.

If we denote the mode shapes of our problem by $\gamma_n(x)$, then the general solution is given by

$$u(x,t) = \sum_{n=1}^{\infty} \gamma_n(x) \cos(\omega_n t + \phi_n) \tag{5.3.24}$$

Equation (5.3.24) is exactly the same form as (5.2.31), and thus we'd use exactly the same approach in matching any initial conditions. Actually, *all* solvable continuous vibration problems will have free responses of this form. The general response will be a summation of the system's eigenmodes, each oscillating at its particular natural frequency. Only the form of the modes will change. For a clamped-clamped beam, we'd still end up with (5.3.24), except for this case the γ's would represent the clamped-clamped modes of the beam, not the free-free ones. If we were looking at plate vibrations, we'd have to solve a plate equation, which is definitely more complex than the beam equation. But we'd still end up with an infinite number of eigenfunctions, each with its own vibration frequency. For this case we'd also have a $\gamma_{m,n}(x, y)$ because the mode shape would have both an x and y dependence, due to the structure being a two-dimensional plate rather than a one-dimensional beam.

5.4 CONTINUOUS SYSTEMS—FORCED VIBRATION

Although we've pretty well covered how to calculate the free vibrations of a continuous system, we haven't yet touched on what we'd do in the face of an external forcing. Just as in the finite-dimensional systems we've already looked at, the response of our systems to an applied forcing will very much depend upon the system's free response characteristics. Thus, having figured out how to get the system's natural modes and frequencies, it's time to turn to the forced problem.

You'll recall that in the finite-dimensional case, applying a force to one of the system's discrete masses excited (in general) all the system modes. A particular eigenmode wasn't excited only when it had a zero in the modal response that was associated with the mass being excited. Otherwise, all modes were excited, and the amplitude of each contributing mode depended upon the amplitude of the forcing and on how close the forcing frequency was to that mode's natural frequency. Well, the same thing occurs with continuous systems. The only thing that's a bit more involved is the case of exciting *only* one particular mode. For the finite-dimensional case we could apply a force to *every* mass and force the system at the correct frequency and amplitudes to excite just a particular mode. Doing this isn't inconceivable if the number of masses isn't too high. But for a continuous system, we'd need an *infinite* number of actuators, since the mode is a distributed one. Clearly this isn't possible. However, it's still possible to get pretty close to just exciting a single mode, if appropriate care is taken.

Let's consider the forced beam problem (all others being similar in approach). We'll rewrite (5.3.7) as follows:

$$\rho(x)\ddot{u} = -(EI(x)u_{xx})_{xx} + f(x,t) \tag{5.4.1}$$

We're generally concerned with sinusoidal forcing and, since we know how to break down any periodic signal into a summation of sinusoids and apply them individually (from Section 3.2), we'll just concern ourselves with a single sinusoidal force,

$$\rho(x)\ddot{u} = -(EI(x)u_{xx})_{xx} + \bar{f}(x)\cos(\omega t) \tag{5.4.2}$$

Now we're going to exploit some of what we know about eigenfunctions. We know from our finite-dimensional work that any possible vector can be represented by an appropriate summation of the system's eigenvectors. We called this spanning the configuration space. The same is true for continuous problems. Thus we shall represent the force distribution over the beam by an infinite summation of the systems's eigensolutions. Therefore, instead of $\bar{f}(x)$, we'll have the following form:

$$\rho(x)\ddot{u} = -(EI(x)u_{xx})_{xx} + \left(\sum_{n=1}^{\infty} \bar{f}_n \gamma_n(x)\right)\cos(\omega t) \tag{5.4.3}$$

In this expression, $\bar{f}_n$ represents the amount of mode $\gamma_n(x)$ that's present in $\bar{f}$. Thus by adding up all these terms, we'll obtain $\bar{f}$.

Just as we did in the finite-dimensional case, we can represent the response as a summation of all the eigenmodes of the system. We'll represent $u(x, t)$ by

$$u(x, t) = \sum_{n=1}^{\infty} a_n(t)\gamma_n \tag{5.4.4}$$

As advertised earlier, the (x) is being dropped to avoid cluttering the integrals that follow. Remember, though, that the γ's are x dependent.

The $a_n(t)$ in (5.4.4) take the role that the generalized coordinates, η, took in our finite-dimensional studies (Section 4.9). What we're saying here is that each eigenmode will contribute to the overall response, and the amount of that contribution will be controlled by each $a_n(t)$.

Substituting (5.4.4) into (5.4.3), we obtain

$$\rho(x)\sum_{n=1}^{\infty} \ddot{a}_n(t)\gamma_n = -\left(EI(x)\sum_{n=1}^{\infty} a_n(t)\gamma_{n_{xx}}\right)_{xx} + \left(\sum_{n=1}^{\infty} \bar{f}_n \gamma_n\right)\cos(\omega t) \tag{5.4.5}$$

Very messy indeed. We've got a single equation that involves *all* the eigenmodes of the system and it doesn't look all that solvable. We need to get a *set* of equations, rather than the summation of terms we're currently looking at. To do this, we'll use the trick we've seen already in Section 4.8, i.e., we'll exploit orthogonality. All the eigenmodes are orthogonal (just as the finite-dimensional ones were). We'll show this mathematically in Section 5.5, but for now you can just take my word for it. To apply orthogonality to this problem, we'll multiply (5.4.5) by one of the eigenmodes (γ_m) and integrate the equation from $x = 0$ to $x = l$:

$$\int_0^l \rho(x)\gamma_m \sum_{n=1}^{\infty} \ddot{a}_n(t)\gamma_n dx = -\int_0^l \gamma_m \left(EI(x)\sum_{n=1}^{\infty} a_n(t)\gamma_{n_{xx}}\right)_{xx} dx + \int_0^l \gamma_m \left(\sum_{n=1}^{\infty} \bar{f}_n \gamma_n\right)\cos(\omega t)dx \tag{5.4.6}$$

Whoa! This looks even worse than before. But is it really? If a system's eigenmodes are all orthogonal, then we should get some simplifications. Again, if you want the details you'll have to wait until the next section. However, if you're content to believe me then you can be amazed when I tell you that almost every term in the above integrals of (5.4.6) goes to zero. Orthogonality for a beam problem means that

$$\int_0^l \rho(x)\gamma_m\gamma_n dx = 0 \tag{5.4.7}$$

and

$$\int_0^l \left(EI(x)\gamma_m\gamma_{n_{xx}}\right)_{xx} dx = 0 \tag{5.4.8}$$

whenever m and n are different. Therefore, only one term out of each integral will remain—the one associated with γ_m. Thus we're left with

$$\begin{aligned}\ddot{a}_m(t)\int_0^l \rho(x)\gamma_m^2 dx = -a_m(t)\int_0^l \gamma_m\left(EI(x)\gamma_{m_{xx}}\right)_{xx} dx \\ + \bar{f}_m\cos(\omega t)\int_0^l \gamma_m^2 dx, \quad m = 1, \ldots, \infty\end{aligned} \tag{5.4.9}$$

For now we won't pursue some of the further simplifications that will be introduced in Section 5.5 and will content ourselves with a few observations. First, you'll note that we have an uncoupled set of equations to solve. Furthermore, it's now eminently solvable. The integral terms may look complicated, but they're just equal to scalar constants. Once you've specified what l, EI, ρ, and the $\gamma_m(x)$'s are, you just integrate and get some result. Thus what we really have is

$$m_m\ddot{a}_m(t) = -k_m a_m(t) + \bar{f}_m\cos(\omega t)c_m, \quad m = 1, \ldots, \infty \tag{5.4.10}$$

c_m, m_m, and k_m are the constants that result from numerically integrating the spatial integrals. Bringing the second term to the left gives us

$$m_m\ddot{a}_m(t) + k_m a_m(t) = c_m\bar{f}_m\cos(\omega t), \quad m = 1, \ldots, \infty \tag{5.4.11}$$

We've *finally* gotten down to something reasonable. All we're left with is the old familiar, forced oscillator problem, and the dependent variable is $a_m(t)$. We've got an equivalent mass (m_m) and an equivalent spring constant (k_m), along with a forcing magnitude ($c_m\bar{f}_m$) and a forcing frequency of ω. So just as in the finite-dimensional case, all we've got to do is determine the response magnitude of each eigenmode due to the applied forcing, multiply by the associated eigensolution, and then sum the solutions up, as shown in (5.4.4).

Example 5.4

Problem Determine the response of a tensioned, uniform, fixed-fixed string to the distributed forcing

$$f = \bar{f}\cos(\omega t)$$

where ω is not equal to any of the string's natural frequencies.

Solution From Appendix A we find the equation of motion of a uniform string to be

$$f(x,t) + Ty_{xx} = \rho\ddot{y}$$

The system is mathematically the same as the fixed-fixed bar examined in Section 5.2, and thus we know that the eigenfunctions are of the form

$$\gamma_n(x) = \sin\left(\frac{n\pi x}{l}\right)$$

Following (5.4.4) we can express the string's displacement as

$$y(x,t) = \sum_{n=1}^{\infty} a_n(t)\gamma_n$$

and from (5.4.9) (using the string equation instead of the bar) we have

$$\rho\ddot{a}_m(t)\int_0^l \gamma_m^2 dx = Ta_m(t)\int_0^l \gamma_{m_{xx}}\gamma_m dx + \bar{f}_m\cos(\omega t)\int_0^l \gamma_m^2 dx, \quad m = 1,\ldots,\infty \tag{5.4.12}$$

A Fourier series decomposition of $\bar{f}$ in terms of the system eigenfunctions yields

$$\bar{f} = \sum_{n=1}^{\infty} \bar{f}_n \sin\left(\frac{n\pi x}{l}\right)$$

$$\bar{f}_n = \frac{4\bar{f}}{n\pi}, \quad n = 1,3,5,\ldots$$

Note that only the odd order harmonics show up in the decomposition.

Using this expansion in (5.4.12) and evaluating the spatial integrals yields

$$\rho\ddot{a}_n + \frac{T(n\pi)^2}{l^2}a_n = \frac{4\bar{f}}{n\pi}\cos(\omega t), \quad n = 1,3,5,\ldots$$

or

$$\ddot{a}_n + \omega_n^2 a_n = f_n\cos(\omega t), \quad n = 1,3,5,\ldots \tag{5.4.13}$$

where $\omega_n^2 = \frac{T(n\pi)^2}{\rho l^2}$ and $f_m = \frac{4\bar{f}}{n\pi\rho}$. The solution to (5.4.13) is

$$a_n(t) = \frac{f_n}{\omega_n^2 - \omega^2}\cos(\omega t)$$

and therefore the complete response is given by

$$y(x,t) = \sum_{n=1,3}^{\infty} \frac{f_n}{\omega_n^2 - \omega^2} \sin\left(\frac{n\pi x}{l}\right) \cos(\omega t)$$

5.5 ORTHOGONALITY OF EIGENFUNCTIONS

As we saw in Section 5.4, the concept of orthogonality exists for continuous systems just as it does for finite-dimensional systems. Having seen that, it is now appropriate to delve a bit more deeply into the concept. To clearly see what's going on with continuous systems, let's review what we've already learned about orthogonality for discrete systems. Back in Section 4.8 we saw that the eigenvectors of a discrete vibration problem usually aren't orthogonal to each other (in a geometric sense) but rather are orthogonal with respect to each other *and* the stiffness or mass matrices, i.e.,

$$X_i^T [M] X_j = 0 \tag{5.5.1}$$

and

$$X_i^T [K] X_j = 0 \tag{5.5.2}$$

where $[M]$ and $[K]$ are the mass and stiffness matrix, respectively, and the X_i's are eigenvectors for the system. Only in the special case for which $[M]$ or $[K]$ is proportional to the identity matrix did we have

$$X_i^T X_j = 0 \tag{5.5.3}$$

Since the mass and stiffness distributions were closely tied up in the concept of orthogonality, it makes sense that they'll still be involved in the continuous case. The only difference is that this time they'll be continuous distributions, rather than lumped quantities.

One of the advantages we had in the finite-dimensional problems was that they all were expressed in the same way:

$$[M]\ddot{X} + [K]X = 0 \tag{5.5.4}$$

where $[M]$ and $[K]$ were $n \times n$ matrices. Therefore once we had demonstrated orthogonality for one $[M]$ and $[K]$ we were done; all other combinations were conceptually the same. Not so, however, for the continuous case. When we deal with continuous systems, we're dealing with partial differential equations, and there are many qualitatively different partial differential equations that apply to vibratory systems. Rather than go to a correct but abstract operator formulation, we'll look at a few basic continuous problems and indicate how any other particular problem can be handled.

We'll start by examining the bar equation:

$$m(x)\ddot{\xi} = (EA(x)\xi_x)_x \tag{5.5.5}$$

If we go through the separation of variables procedure, we can find the natural frequencies ω_i and eigenfunctions $\bar{\xi}_i$ and, substituting them into (5.5.5), obtain

$$-m(x)\omega_i^2\bar{\xi}_i = \frac{\partial}{\partial x}\left(EA(x)\frac{\partial\bar{\xi}_i}{\partial x}\right) \tag{5.5.6}$$

This equation also holds for $\bar{\xi}_j$, the jth eigenfunction

$$-m(x)\omega_j^2\bar{\xi}_j = \frac{\partial}{\partial x}\left(EA(x)\frac{\partial\bar{\xi}_j}{\partial x}\right) \tag{5.5.7}$$

We'll now multiply (5.5.6) by $\bar{\xi}_j$ and integrate along the beam while also multiplying (5.5.7) by $\bar{\xi}_i$ and integrating along the beam as well, giving us

$$-\int_0^l m(x)\omega_i^2\bar{\xi}_j\bar{\xi}_i\,dx = \int_0^l \bar{\xi}_j\frac{\partial}{\partial x}\left(EA(x)\frac{\partial\bar{\xi}_i}{\partial x}\right)dx \tag{5.5.8}$$

and

$$-\int_0^l m(x)\omega_j^2\bar{\xi}_i\bar{\xi}_j\,dx = \int_0^l \bar{\xi}_i\frac{\partial}{\partial x}\left(EA(x)\frac{\partial\bar{\xi}_j}{\partial x}\right)dx \tag{5.5.9}$$

Next, we want to subtract (5.5.9) from (5.5.8):

$$\begin{aligned}(\omega_j^2-\omega_i^2)\int_0^l m(x)\bar{\xi}_i\bar{\xi}_j dx &= \int_0^l \bar{\xi}_j\frac{\partial}{\partial x}\left(EA(x)\frac{\partial\bar{\xi}_i}{\partial x}\right)dx \\ &\quad -\int_0^l \bar{\xi}_i\frac{\partial}{\partial x}\left(EA(x)\frac{\partial\bar{\xi}_j}{\partial x}\right)dx\end{aligned} \tag{5.5.10}$$

Just to keep a grip on where we're going, refer back to the finite-dimensional work. We did a similar thing back in Section 4.8 when we premultiplied the equations by X_j^T and X_i^T and then subtracted them. We ended up with a quadratic that involved the $[M]$ matrix, which was multiplied by the difference of two natural frequencies ($\omega_2^2-\omega_1^2$ for the specific example we were examining). It then turned out that the right-hand side (in that case the difference between two quadratic expressions involving $[K]$) was equal to zero. Thus we found the orthogonality relation that $X_i^T[M]X_j = 0$ for $i \neq j$.

We're going to find just about the same result here. The left-hand side is a quadratic relation between ξ_i, ξ_j, and the mass distribution $m(x)$. It's multiplied by the difference between two natural frequencies. All that's left is to show that the right-hand side equals zero. To do this, we'll use a mathematical operation that we'll be using quite a bit in Chapter 6: integration by parts. When we integrate by parts, we use the fact that

$$\int_0^l u\,dv = uv\Big|_0^l - \int_0^l v\,du \tag{5.5.11}$$

Our right-hand sides are just begging to be integrated by parts. For instance, if we look at the first term after the equality sign in (5.5.10) we see that identifying $\bar{\xi}_j$ as u and $\frac{\partial}{\partial x}\left(EA(x)\frac{\partial\bar{\xi}_i}{\partial x}\right)dx$ as dv

allows us rewrite it as

$$\int_0^l \bar{\xi}_j \frac{\partial}{\partial x}\left(EA(x)\frac{\partial \bar{\xi}_i}{\partial x}\right)dx = \bar{\xi}_j\left(EA(x)\frac{\partial \bar{\xi}_i}{\partial x}\right)\Bigg|_0^l - \int_0^l EA(x)\frac{\partial \bar{\xi}_i}{\partial x}\frac{\partial \bar{\xi}_j}{\partial x}dx \tag{5.5.12}$$

In the same manner, the last term on the right-hand side of (5.5.10) can be written as

$$\int_0^l \bar{\xi}_i \frac{\partial}{\partial x}\left(EA(x)\frac{\partial \bar{\xi}_j}{\partial x}\right)dx = \bar{\xi}_i\left(EA(x)\frac{\partial \bar{\xi}_j}{\partial x}\right)\Bigg|_0^l - \int_0^l EA(x)\frac{\partial \bar{\xi}_i}{\partial x}\frac{\partial \bar{\xi}_j}{\partial x}dx \tag{5.5.13}$$

If we substitute both these results into (5.5.10), we'll obtain

$$(\omega_j^2 - \omega_i^2)\int_0^l m(x)\bar{\xi}_i\bar{\xi}_j\,dx = \bar{\xi}_j\left(EA(x)\frac{\partial \bar{\xi}_i}{\partial x}\right)\Bigg|_0^l - \bar{\xi}_i\left(EA(x)\frac{\partial \bar{\xi}_j}{\partial x}\right)\Bigg|_0^l \tag{5.5.14}$$

The integrals terms are identical, thus they canceled each other out, leaving us with just the boundary condition terms. Well, what if the beam is fixed at both ends? Then $\bar{\xi}_i$ and $\bar{\xi}_j$ are both zero, thus making the right-hand side zero. And what if the beam is fixed at one end and free at the other? Then the displacement is zero at one end and the slope is zero at the other, and the boundary terms again go to zero. In fact, you can put in whatever combination of boundary terms you'd like. When you evaluate the right-hand side, you'll see that both terms either are zero or exactly cancel each other. Therefore we're left with what we were hoping for, namely,

$$(\omega_j^2 - \omega_i^2)\int_0^l m(x)\bar{\xi}_i\bar{\xi}_j\,dx = 0 \tag{5.5.15}$$

As in the discrete case, we realize that if all the natural frequencies are different, then $\omega_i^2 - \omega_j^2 \neq 0$ and therefore

$$\int_0^l m(x)\bar{\xi}_i\bar{\xi}_j dx = 0 \tag{5.5.16}$$

This is our continuous bar analogue to $X_i^T[M]X_j = 0$. This orthogonality condition doesn't involve the eigenfunctions alone but also includes the mass distribution, just as it did in the discrete case. And whereas the discrete case involved vector/matrix multiplications, here we have multiplications of the functions followed by integration over the length of the bar. Notice that if $m(x)$ is constant, then it factors out of the integral and we're left with

$$\int_0^l \bar{\xi}_i\bar{\xi}_j dx = 0 \tag{5.5.17}$$

In this case the functions are actually orthogonal to each other, just as the discrete vectors X_i are for mass matrices, which are just constant multiples of the identity matrix.

Finally, using this orthogonality relation in (5.5.9) gives us our other orthogonality relation:

$$\int_0^l \bar{\xi}_i \frac{\partial}{\partial x}\left(EA(x)\frac{\partial \bar{\xi}_j}{\partial x}\right)dx = 0 \tag{5.5.18}$$

5.6 HOMEWORK PROBLEMS

Note: Assume constant material properties for all the following problems.

Section 5.2

5.1. Determine the boundary conditions for the system illustrated in Figure P5.1. T is the tension in the string and ρ is the linear mass density.

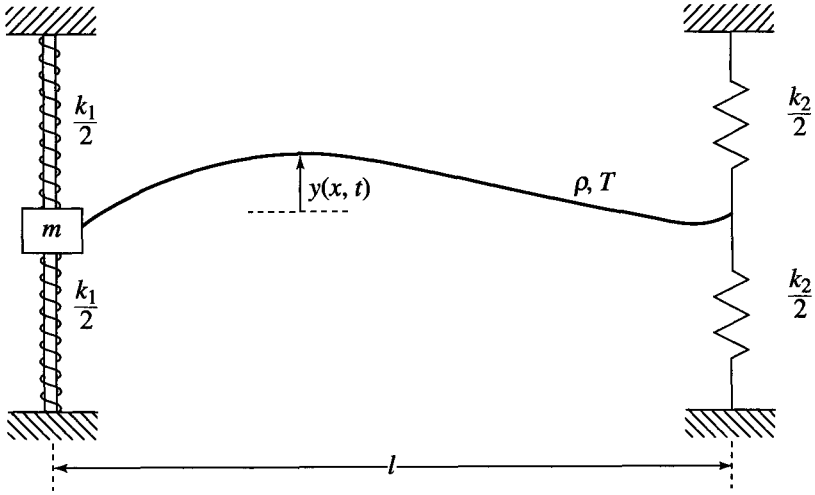


Figure P5.1

5.2. Find the natural frequencies and eigenfunctions for the, spring-restrained string shown in Figure P5.2. Let $k_1 = 90$ N/m, $k_2 = 150$ N/m, $T = 1500$ N, $l = 1$ m, and $\rho = .1$ kg/m. Show that the frequencies go to $\frac{n\pi}{l}\sqrt{\frac{T}{\rho}}$ for $k_1 = k_2 = 0$ and $k_1 = k_2 = \infty$ and to $\left(\frac{2n-1}{2}\right)\frac{\pi}{l}\sqrt{\frac{T}{p}}$ for $k_1 = 0$, $k_2 = \infty$ and $k_2 = \infty, k_1 = 0$.

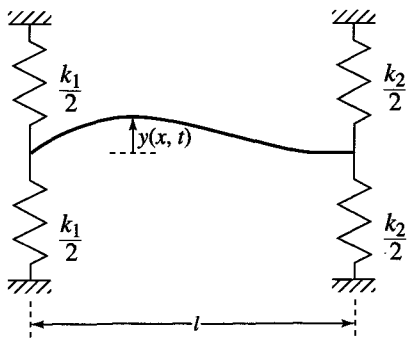


Figure P5.2

5.3. Assume that you wish to use a noncontacting sensor to observe the odd modes of oscillation of a string. The string has uniform mass and tension, is fixed at both ends, and is 1 m long. Would it be better to place the sensor near $x = .5$ m or near $x = \frac{\sqrt{2}}{2}$ m? Why?

5.4. Figures P5.4 shows two snapshots in time of a tensioned string undergoing free vibration. Explain how such a situation could arise if the system modes were proportionally damped. Explain how it could also occur under nonproportional damping.

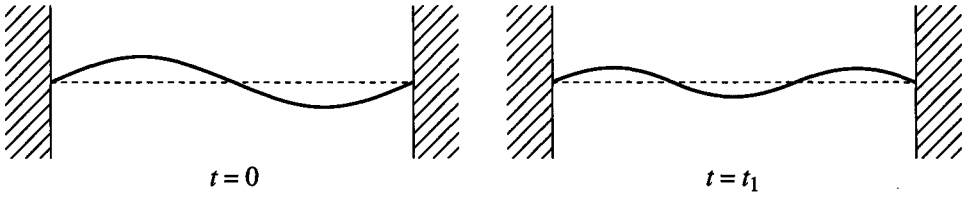


Figure P5.4

5.5. When discussing lumped spring responses in Chapter 1 it was mentioned that the exact solution for the first natural frequency for the system illustrated in Figure P5.5 was .9836 rad/s. Verify this fact and show that for high frequencies, the natural frequencies, tend toward $\pi\sqrt{\frac{EA}{m}}$. $m = 1$ kg/m, $EA = 10$ N, $m_1 = 10$ kg, $l = 1$ m.

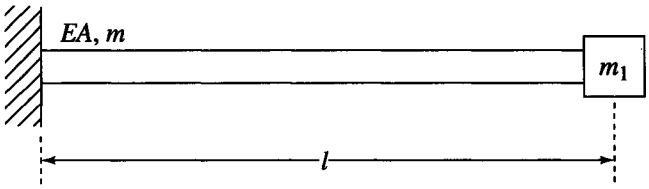


Figure P5.5

5.6. By how much will the pitch of a note change if the density of the corresponding guitar string is halved and the tension is doubled?

5.7. The G string of a violin has a mass of .25 g and a length of 38 cm. What is the tension needed to get the string to the proper pitch? (Assume that the G below middle C occurs at 196 Hz.) How many pounds is this equivalent to?

5.8. If you increase the tension of a guitar string by 10%, by how much does the pitch alter?

5.9. It is often difficult to get a really good clamped boundary condition experimentally. Assume that insufficient clamping conditions have caused the effective beam length to go to $l + dl$, rather than the l you planned for in Figure P5.9. How much will your natural frequencies be altered by? Can you use this shift as a tool for determining whether your beam is clamped properly?

5.10. Solve for the eigenfunctions and eigenfrequencies of a uniformly tensioned string for which the boundary conditions are $w_x(0) = w_x(l) = 0$, i.e., zero slope at each end.

5.11. Is it possible for a tensioned, spring-restrained string, oscillating at one of its natural frequencies, to have the modal deflection illustrated in Figure P5.11?

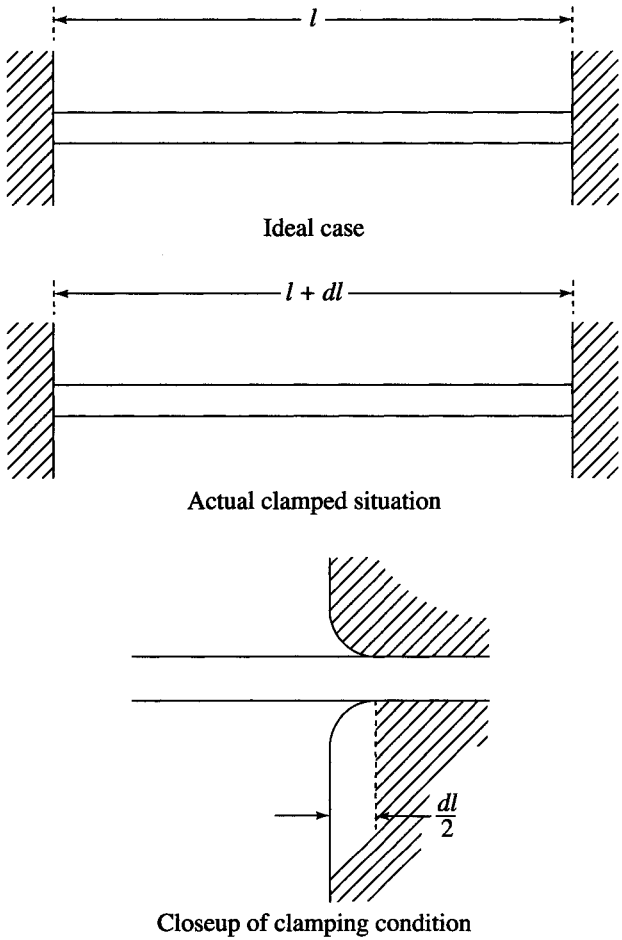


Figure P5.9

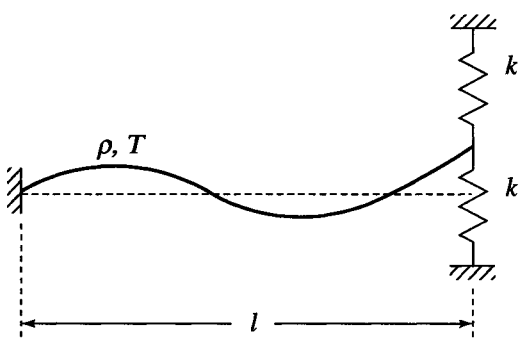


Figure P5.11

5.12. The natural frequencies of the system illustrated in Figure P5.12*a* are displayed versus a system parameter in Figure P5.12*b*, *c*. What are the possible parameters that might reasonably appear on the x axis of each plot? Why?

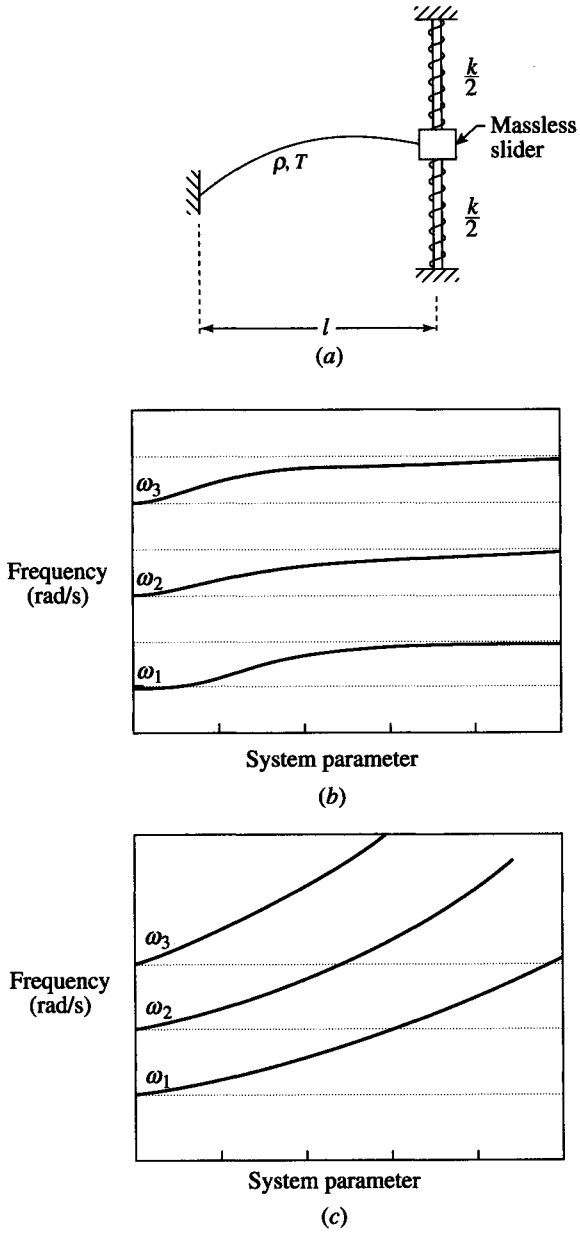


Figure P5.12

5.13. Determine the correct boundary conditions for the system shown in Figure P5.12*a* and indicate graphically where the β solutions lie.

5.14. Derive the β equation that will let you solve for the β values of the system shown in Figure P5.14. Consider longitudinal motions of the bar/lumped masses.

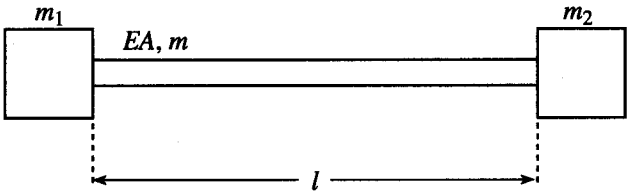


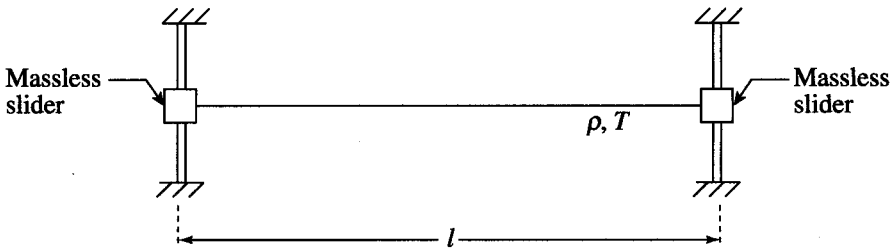
Figure P5.14

5.15. Determine the correct boundary conditions for the system shown in Figure P5.5 and indicate graphically where the β solutions lie.

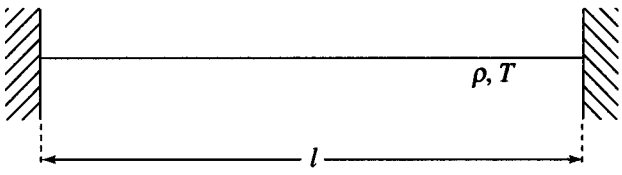
5.16. Let's look more closely at Problem 5.13. Let $l = 1$ m, $T = 100$ N, and $\rho = .002$ kg/m. Show that the β solutions move from those of a restrained-free string $\left(\left(\frac{2n-1}{2}\right)\pi\right)$ to those of a restrained-restrained string $(n\pi)$, as k is increased from 0 toward ∞.

5.17. We'll look more closely at Problem 5.15 in this exercise. Show that as m_1 varies from 0 toward ∞, the $\beta_i l$ values move from $\left(\frac{2n-1}{2}\right)\pi$ to $n\pi$, i.e., from the fixed-free to the fixed-fixed solutions. Let $m = .2$ kg/m, $l = 2$ m, and $EA = 1000$ N.

5.18. Show that the β values for a vertically unrestrained, tensioned string are the same as those for a fully restrained, tensioned string (Figure P5.18). Are the eigenfunctions also identical?



Unrestrained tensioned string



Fixed-fixed tensioned string

Figure P5.18

5.19. Show how to calculate the natural frequencies of a tensioned string (Figure P5.19) for which the linear mass density is constant at ρ_0 from $x = 0$ to $x = l_1$ and constant at $\rho = \rho_1$ from $x = l_1$ to $x = l$.

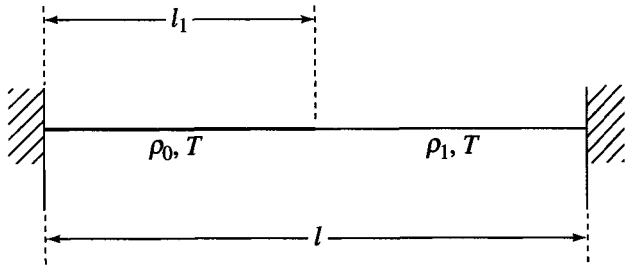


Figure P5.19

5.20. What is the response of a fixed-fixed tensioned string to the following initial conditions?

$$y(x) = \sin\left(\frac{2\pi x}{l}\right) \quad \text{at } t = 0$$

$$\dot{y}(x) = \sin\left(\frac{\pi x}{l}\right) \quad \text{at } t = 0$$

5.21. What is the free response of a fixed-free bar given the following initial conditions?

$$\xi(x, 0) = ax \quad \text{at } t = 0$$

$$\dot{\xi}(x, 0) = 0 \quad \text{at } t = 0$$

5.22. Consider Problem 5.19. Show that if $\rho_0 = \rho_1$, you obtain the expected uniform, tensioned string results.

5.23. Consider Problem 5.19. Show that if ρ_0 is much greater than ρ_1 you'll obtain natural frequencies corresponding to vibrations of a string having tension T, density ρ_1, and length $l - l_1$. Explain why such a result occurs. Show also that natural frequencies

$$\omega_n = \sqrt{\frac{T}{\rho_0}}\frac{(2n-1)\pi}{2l_1}$$

are obtained, and explain what modes they correspond to.

5.24. Determine the β equation for the system shown in Figure P5.24 (i.e., the equation that lets you find the β_n for the eigenfunctions).

5.25. Determine the response of a fixed-fixed bar to the following initial conditions:

$$\xi(x, 0) = 0$$

$$\dot{\xi}(x, 0) = b\sin\left(\frac{3\pi x}{l}\right)$$

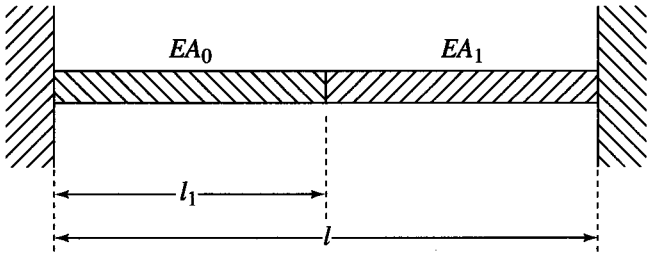


Figure P5.24

5.26. Calculate and plot the response at $x = .5\,l$ and $.75\,l$ for a pinned-pinned string having the following initial conditions:

$$y(x, 0) = 0$$

$$\dot{y}(x, 0) = \sin\left(\frac{\pi x}{l}\right) - \sin\left(\frac{2\pi x}{l}\right)$$

that is, solve for the particular values of the general response

$$y(x, t) = \sum_{n-1}^{N} a_n \sin(\beta_n x)\cos(\omega_n t - \phi_n)$$

Plot the responses for $N = 1$ and $N = 2$, i.e., a one- and two-term approximation. Comment on any differences between the two cases.

5.27. If the fundamental frequency of a fixed-fixed tensioned string should equal 440 Hz (A above middle C) and the string is 1 m long and has a mass of .009 kg, what must the tension be?

5.28. How do the natural frequencies differ for a 1 m long bar made of steel, aluminum, and copper? Assume fixed-free boundary conditions.

5.29. A bar is thrown into a rigid wall as shown below in Figure P5.29. It contacts the wall with a speed of v m/s at $t = 0$. Determine the response of the bar for $t > 0$. Assume that the right end of the bar remains attached to the wall. $l = 1$ m.

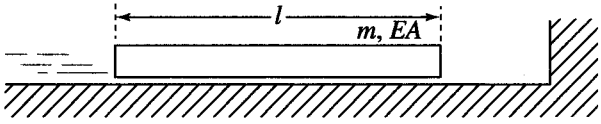


Figure P5.29

5.30. A tensioned string is struck in an impulsive manner at $t = 0$ and $x = .5l$. The initial conditions resulting from the impact are given by

$$y(x) = 0 \quad \text{at } t = 0$$

$$\dot{y}(x) = a\delta(x - .5l) \quad \text{at } t = 0$$

Determine the string's response for $t > 0$.

5.31. Determine the first four natural frequencies for the system illustrated in Figure P5.31. $\rho = 9.2 \times 10^3$ kg/m^3, $G = 8.5 \times 10^{10}$ N/m^2, $J = 1$ m^4, $l = 1.5$ m.

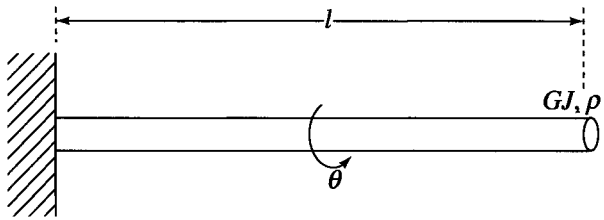


Figure P5.31

5.32. Determine the percentage change in the first three natural frequencies for torsional motion if a circular disk is attached to the end of the circular rod shown in Figure P5.32. $I_0 = .2$ kg· m^2, $l = 1$ m, $G = 7.3 \times 10^{11}$ N/m^2, $\rho = 7.2 \times 10^3$ kg/m^3, $J = 4 \times 10^{-7}$ m^4.

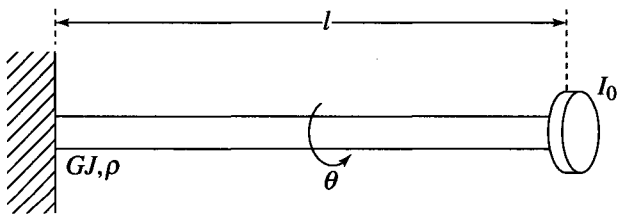


Figure P5.32

5.33. Figure P5.33 below shows part of a machine in which a torsional spring-restrained, circular shaft is moved by a sinusoidally varying torque applied at x_1. At what frequency of excitation will the system stop being well approximated as a rigid shaft in a spring restraint? $l = 1$ m, $G = 6.2 \times 10^{11}$ N/m^2, $\rho = 6.4 \times 10^3$ kg/m^3, $J = 2 \times 10^{-7}$ m^4, $k_\theta = 1.5 \times 10^5$ N·m/rad.

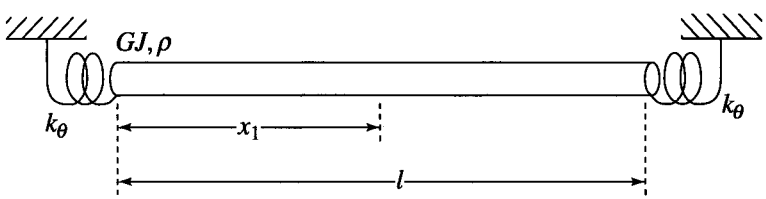


Figure P5.33

5.34. In Problem 5.33 we looked at a spring-restrained circular shaft. Numerically determine the lowest frequency oscillation mode, and show that it matches the prediction you'd obtain from a lumped analysis.

5.35. Consider a fixed-free circular rod made of steel. What is lower, the first torsional or the first longitudinal natural frequency? $G = 7.9 \times 10^{10}$ N/m^2, $E = 2.1 \times 10^{11}$ N/m^2, $\rho = 7.8 \times 10^3$ kg/m^2.

5.36. Bicycle designers worry about lightness and stiffness (among other concerns) when designing their frames. Determine which is torsionally stiffer, a smaller diameter tube or a larger one. Use a constant mass of metal each time. The first tube will have an inside diameter of 3 cm and a wall thickness of .5 mm. The second tube will have an inside diameter of 4 cm. We'll determine stiffness by calculating the first natural frequency of a fixed-free tube upon which we've attached a lumped inertia I_0. $G = 7.9 \times 10^{10}$ N/m^2, $\rho = 7.8 \times 10^3$ kg/m^3, $l = 1$ m, $I_0 = 5 \times 10^{-3}$ kg·m^2.

5.37. Assume that you're employed as a bicycle designer and your task is to oversee the move from steel to aluminum. Your task is to decrease the tubing's weight by 10% and at the same time increase its torsional stiffness. Consider a fixed-fixed tube, 1 m long. The wall thickness of the steel tube is .5 mm, and the inside radius is 12.5 mm.

(a) Calculate the natural frequencies of both a steel tube and a tube of the same dimensions made of aluminum.

(b) Calculate the torsional stiffness for each tube from part (a) (torsional stiffness $= GJ$).

(c) Increase the inside radius of the aluminum tube by 30% and determine what the wall thickness must be to attain a 10% weight reduction over the steel tube.

(d) How much stiffer is the tube designed in part (c) than the steel tube?

5.38. Is there any difference in the torsional natural frequencies of a solid circular rod and a hollow circular shaft, assuming that they have the same length and mass? Consider the case of fixed-free boundary conditions.

5.39. Consider again the system of Figure P5.33. Let the torsional spring on the left end be infinitely stiff. How large must k_θ be for the first torsional natural frequency of this system to be within 5% of the first torsional natural frequency for fixed-fixed conditions? $l = 1$ m, $G = 6 \times 10^{10}$ N/m^2, $J = 1 \times 10^{-8}$ m^4, $\rho = 7 \times 10^3$ kg/m^3.

5.40. Putting a lumped rotational inertia on the end of a fixed-free, hollow, circular shaft will produce natural frequencies lower than those of the original system. How large a rotational inertia can be used before the first frequency shifts by 10%? The inner radius of the shaft is 1 cm, the wall thickness is 1 mm, the length is 2 m, $G = 5.7 \times 10^{10}$ N/m^2, and $\rho = 6.7 \times 10^3$ kg/m^3.

5.41. Consider again the system of Problem 5.12. How large a spring constant k should be used to bring the first natural frequency to 185 rad/s from its value when $k = 0$? $l = 2$ m, $T = 1000$ N, $\rho = .02$ kg/m.

5.42. The system in Figure P5.42 has natural frequencies that are essentially equal to the fixed-fixed ones, as well as a low frequency, "rigid body" mode. How small can m_1 get before the second natural frequency has increased to 10% above its original value? $l = 3$m, $\rho = .02$ kg/m, $T = 3000$ N, $m_1 = 1000$ kg.

5.43. The system illustrated in Figure P5.43 has natural frequencies of $\omega_1 = 1000$ rad/s, $\omega_2 = 3000$ rad/s, $\omega_3 = 5000$ rad/s, ... for $m_1 = 0$. As m_1 is increased, the frequencies drop. Can a value of m_1 be found such that $\omega_i = 2100$ rad/s and $\omega_{i+1} = 2700$ rad/s, for some i? Why or why not?

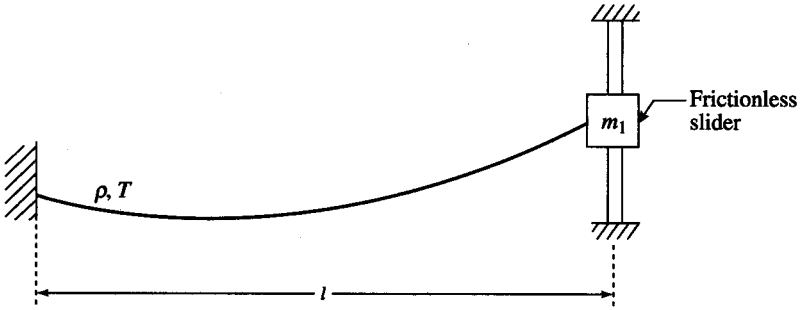


Figure P5.42

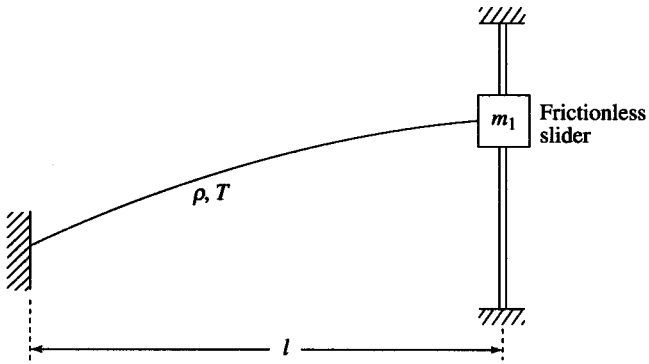


Figure P5.43

5.44. What are the first three natural frequencies for the system illustrated in Figure P5.44 (longitudinal motion)? $EA = 2 \times 10^7$ N/m, $\rho = 3 \times 10^3$ kg/m^3, $l = 1$ m, $A = 1 \times 10^{-4}$ m^2, $k = 5 \times 10^5$ N/m.

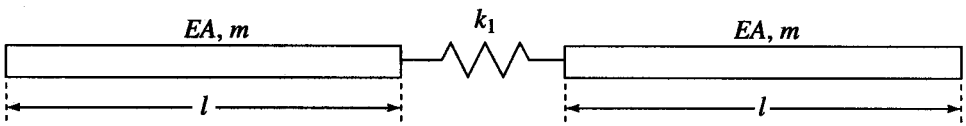


Figure P5.44

5.45. Treating the system illustrated in Figure P5.45 as two identical bars joined by a lumped spring, determine the first two β values for the system. $EA = 2 \times 10^7$ N/m, $\rho = 3 \times 10^3$ kg/m^3, $l = 1$ m, $A = 1 \times 10^{-4}$m^2, $k = 5 \times 10^5$ N/m.

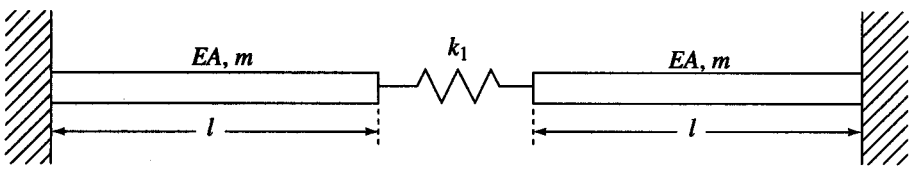
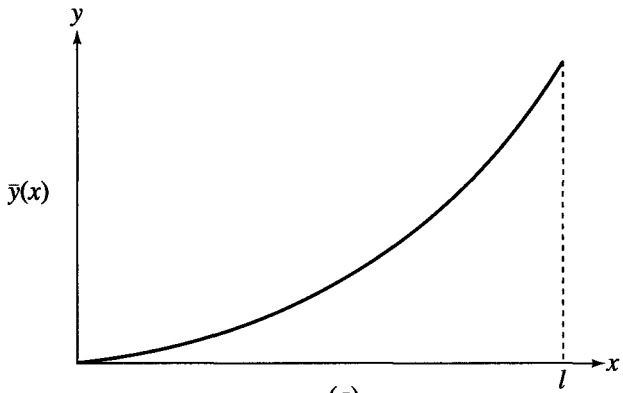


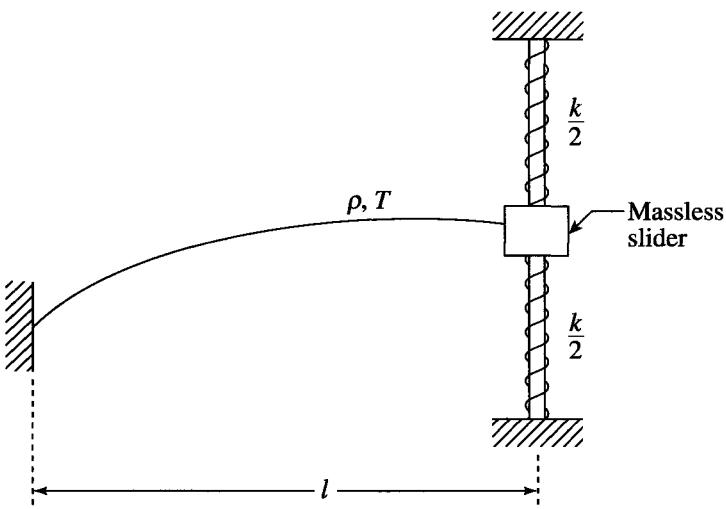
Figure P5.45

5.46. Show analytically that the odd harmonics (1st, 3rd, 5th, etc.) of the lowest natural frequency for a fixed-fixed string of the length l are exactly equal to the natural frequencies of a string of length $l/2$ that is both fixed at one end and vertically unrestrained at the other. Explain why this is the case.

5.47. Why can't the modal response illustrated in Figure P5.47*a* exist as a free vibration eigenfunction of a fixed-spring restrained tensioned string as in Figure P5.47*b*.



(*a*)



(*b*)

Figure P5.47

5.48. In Figure P5.48, can you predict any of the natural frequencies for system A, given the natural frequencies of system B?

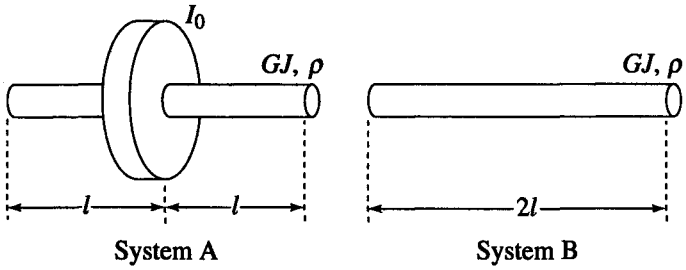


Figure P5.48

5.49. What is the largest change in the natural frequencies that you'd expect if to the original tensioned string of Figure P5.49*a* you added a spring restraint as in Figure P5.49*b*?

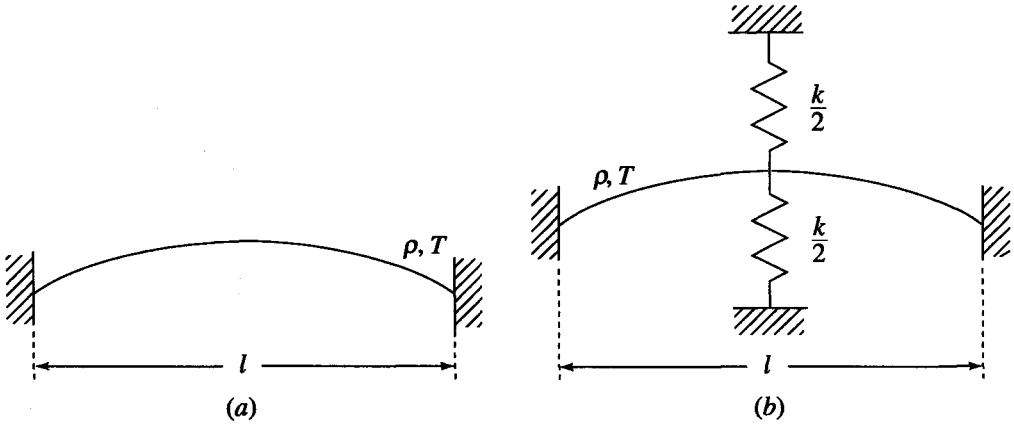


Figure P5.49

5.50. Solve for the eigenfunctions of a fixed-free bar. Use this information to find the strain (first spatial derivative with respect to x). Where is the strain a maximum for the eigenfunction? Where is it a minimum? How would you use this information if you were using strain gauges as sensors?

Section 5.3

5.51. Show that the β values (hence the natural frequencies) are the same for both a free-free beam and a clamped-clamped beam.

5.52. Determine the first four βl values for a beam like that in Figure P5.52: pinned on one end and both pinned and torsionally spring restrained at the other end.

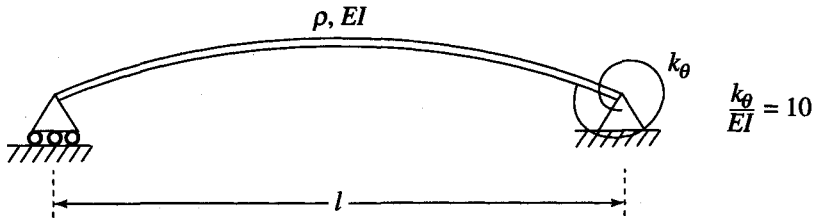


Figure P5.52

5.53. Figure P5.53 shows a pinned-pinned beam on a resilient foundation. Calculate the natural frequencies and eigenfunctions. The stiffness per unit length of the foundation, k, is 175 N/m/m, $l = 2$ m, $EI = 22.4$ N/m^2, $\rho = .624$ kg/m.

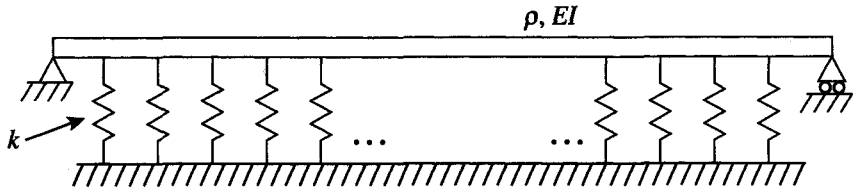


Figure P5.53

5.54. What is the equation of motion for free vibration of a beam with nonconstant stiffness and nonconstant mass distribution?

5.55. Often the boundary conditions for a beam are not completely fixed, and thus an accurate model should include discrete torsional spring restraints at the ends, as shown in Figure P5.55. Derive an expression that allows you to solve for the system's β values. There's no need to obtain a single equation; leaving the problem in matrix form is sufficient.

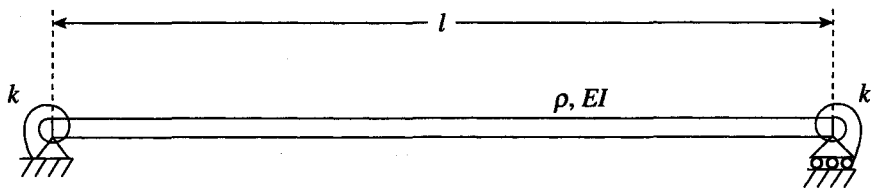


Figure P5.55

5.56. Derive the equation of motion for a beam under tension, T, as in Figure P5.56. Assume constant T, EI, and ρ. Then derive the equation needed to solve for the beam's natural frequencies.

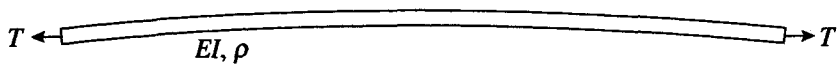


Figure P5.56

5.57. The eigenfunctions for a fixed-fixed bar were given by $\sin(n\pi x)$ and the natural frequencies were evenly spaced: $\omega_n = \beta_n\sqrt{\frac{EA}{m}}$, where $\beta_n = n\pi$. Show that the eigenfunctions for a pinned-pinned beam are also given by $\sin(n\pi x)$. Are the natural frequencies also evenly spaced?

5.58. What sort of beam deflection would you have if the displacement, slope, moment, and shear were zero at $x = 0$, with no conditions specified at $x = l$?

5.59. Derive another (equivalent) form for the mode shape of the beam in Example 5.3.

Section 5.4

5.60. Calculate the response of a fixed-free bar if the applied forcing is of the form

$$f(x,t) = \bar{f}\sin\left(\frac{3\pi x}{2l}\right)\sin(\omega t)$$

for which ω is not equal to any of the bar's natural frequencies.

5.61. Determine the forced response of a tensioned string to an applied forcing of the form shown in the following equation. ω isn't equal to any of the string natural frequencies. The boundary conditions of the string are given by $y(0,t) = y(l,t) = 0$.

$$f(x,t) = \bar{f}\delta(x - x_0)\cos(\omega t)$$

5.62. What is the forced response of a fixed-free circular rod subjected to a time varying torque at $x = x_1$?

$$\tau = \bar{\tau}\delta(x - x_1)\cos(\omega t)$$

5.63. Determine the general response of a pinned-pinned beam when subjected to a distributed force of the form

$$f(x) = 3\sin(\pi x)\sin(5t) - 2\sin(2\pi x)\sin(6t).$$

Assume constant beam properties and let $l = 1$ m.

5.64. What is the response of a pinned-pinned uniform beam to a concentrated, time varying load $f(t)$ applied to the beam's midpoint? Assume that ω_f doesn't correspond to any of the natural frequencies of the system in Figure P5.64.

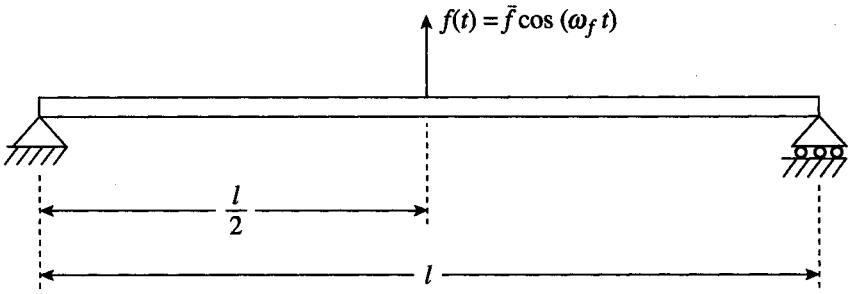


Figure P5.64

5.65. What will the response be of a free-free beam (length l, mass per unit length ρ, bending stiffness EI) to a transverse distributed force of $\bar{f}$ N/m? Note that the force is constant, both in time and with respect to x.

Section 5.5

5.66. Show that the eigenfunctions of a vertically unrestrained, tensioned string are orthogonal to each other.

5.67. Show that the eigenfunctions of a string that's fixed at one end and vertically unrestrained at the other are mutually orthogonal.

5.68. Show that the eigenfunctions of a uniform, unrestrained torsional rod are orthogonal. Ignore the zero-frequency, rigid body solution.

5.69. Numerically solve for two different eigenfunctions for the system shown in Figure P5.69, and demonstrate numerically that they satisfy orthogonality. $l = 1.0$ m, $EA = 2.0 \times 10^7$ N/m, $m = 0.8$ kg, $k = 5.0 \times 10^5$ N/m, $m_1 = .3$ kg

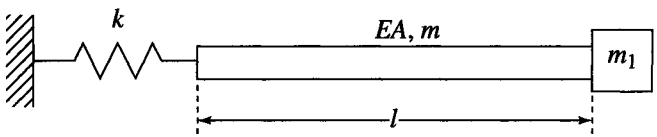


Figure P5.69

5.70. Numerically calculate the first two eigenfunctions of a fixed-fixed bar and verify that they satisfy orthogonality. $EA = 8.0 \times 10^6$ N, $m = 2.0$ kg/m, $l = 1.0$ m.

5.71. Demonstrate that the eigenfunctions of the fixed-free, uniform circular rod in Figure P5.71 satisfy orthogonality.

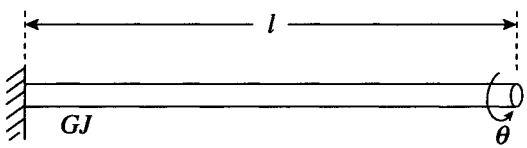


Figure P5.71

5.72. Consider the system of Figure P5.69. Set m_1 equal to zero. Numerically calculate the first two eigenmodes and demonstrate that they satisfy orthogonality. $EA = 5.0 \times 10^6$ N, $k = 7.787 \times 10^6$ N/m, $l = 1.0$ m, $m = 1.0$ kg/m.

5.73. Consider the system shown in Figure P5.5. Show that the first two numerically calculated modes satisfy orthogonality (i.e., generate the modes numerically (vector representation) and numerically evaluate $\int_0^l m\phi_i\phi_j dx + m_1\phi_i(1)\phi_j(1))$. $m = 1$ kg/m, $l = 1$ m, $EA = 1.3 \times 10^6$ N, $m_1 = 1.5$ kg.

5.74. Show that the first two modes of the system illustrated in Figure P5.74 satisfy orthogonality. (Note that the first mode is a rigid body mode.) Numerically calculate the β's and the eigenfunctions and numerically

verify orthogonality. $G = 7 \times 10^9$ N/m^2, $J = 2 \times 10^{-3}$ m^4, $\rho = 3 \times 10^3$ kg/m^3, $l = 1$ m, $I_1 = 12$ kg·m^2, $I_2 = 12$ kg·m^2.

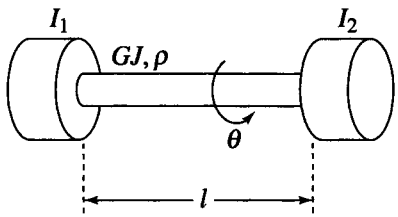


Figure P5.74

5.75. Consider again the system of Figure P5.74. Set I_2 equal to zero. Show that orthogonality holds by numerically generating the first two eigenfunctions for the system and then numerically evaluate

$$\int_0^l \rho J \phi_i(x)\phi_j(x)dx + I_1\phi_i(0)\phi_j(0)$$

$G = 7 \times 10^9$ N/m^2, $J = 2 \times 10^{-3}$ m^4, $\rho = 3 \times 10^3$ kg/m^3, $l = 1$ m, $I_1 = 12$ kg $\cdot$ m^2.